

5G RAN

mmWave Beam Management (High-Frequency TDD) Feature Parameter Description

Issue	Draft A
Date	2025-12-31



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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the capacity scenario with 2:3 slot configuration by the mmWave 3D coverage pattern. For details, see: • 5.1 Principles • 5.3.3 Hardware • 5.3.4 Cells	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for PRACH configuration index 188 when the 2:3 slot configuration is used in the default scenario and customized beam scenario. For details, see 4.3.4 Cells and 5.3.4 Cells .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added support for LampSite mmWave scenario-based beam in mmWave beam management. For details, see: • 2.3 Differences Between NSA and SA • 2.4 Differences Between High Frequency Bands and Low Frequency Bands • 3 Overview • 4.1 Principles • 4.3.3 Hardware • 6 LampSite mmWave Scenario-based Beam • 7.3.3 Hardware	None	High-frequency TDD	DBS5900 LampSite
Added a mutually exclusive relationship between the single-beam scenario of mmWave 3D coverage pattern and intelligent beam selection. For details, see 5.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship between the single-beam scenario of mmWave 3D coverage pattern and PUCCH receive channel adaptation. For details, see 5.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added mutually exclusive relationships between mmWave PUSCH MU-MIMO and the following scenarios: the single-beam scenario and customized beam scenario of mmWave 3D coverage pattern. For details, see 5.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship between the wide-beam scenario of mmWave 3D coverage pattern and mmWave PUSCH MU-MIMO. For details, see 5.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship between Flexible Dense Beam and mmWave PUSCH MU-MIMO. For details, see 8.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for the 2:3 slot configuration in mmWave beam management. For details, see: 4.1 Principles 4.3.4 Cells 5.3.4 Cells 8.3.4 Cells	None	High-frequency TDD	5900 series base stations (macro base stations)

Editorial Changes

Added a figure illustrating the default scenario with 2:3 slot configuration (2_3_DSUUU). For details, see 5.1 Principles.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010015	Basic Beam Management	4 mmWave Basic Beam Management
FOFD-030201	mmWave 3D Coverage Pattern	5 mmWave 3D Coverage Pattern
FOFD-030201	mmWave 3D Coverage Pattern	7 TA Command Delivery at a Short Interval
FOFD-030201	mmWave 3D Coverage Pattern	8 Flexible Dense Beam
FOFD-030201	mmWave 3D Coverage Pattern	9 Dynamic Beam Allocation

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
mmWave basic beam management	None	4 mmWave Basic Beam Management
mmWave 3D coverage pattern	None	5 mmWave 3D Coverage Pattern
LampSite mmWave scenario-based beam management	None	6 LampSite mmWave Scenario-based Beam
TA command delivery at a short interval	None	7 TA Command Delivery at a Short Interval
Flexible Dense Beam	None	8 Flexible Dense Beam
Dynamic beam allocation	None	9 Dynamic Beam Allocation
Optimal selection of narrow beams for access channels	None	10 Optimal Selection of Narrow Beams for Access Channels
Antenna large physical downtilt mode	None	11 Antenna Large Physical Downtilt Mode

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

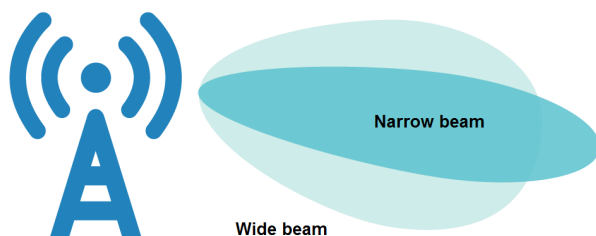
Function Name	Difference	Chapter/Section
mmWave basic beam management	Supported only in high frequency bands In low frequency bands, digital beams are used, where both adaptive weights and fixed weights are used for beamforming. In high frequency bands, analog beams are used, where only fixed weights are used for beamforming. This document describes beam management in high frequency bands (mmWave). For beam management in low frequency bands, see <i>Beam Management</i> .	4 mmWave Basic Beam Management
mmWave 3D coverage pattern	Supported only in high frequency bands	5 mmWave 3D Coverage Pattern
LampSite mmWave scenario-based beam management	Supported only in high frequency bands	6 LampSite mmWave Scenario-based Beam
TA command delivery at a short interval	Supported only in high frequency bands	7 TA Command Delivery at a Short Interval
Flexible Dense Beam	Supported only in high frequency bands	8 Flexible Dense Beam
Dynamic beam allocation	Supported only in high frequency bands	9 Dynamic Beam Allocation
Optimal selection of narrow beams for access channels	Supported only in high frequency bands	10 Optimal Selection of Narrow Beams for Access Channels

Function Name	Difference	Chapter/Section
Antenna large physical downtilt mode	Supported only in high frequency bands	11 Antenna Large Physical Downtilt Mode

3 Overview

The New Radio (NR) system uses the beamforming technology to allow all types of channels and signals to use narrow beams with higher energy density and better directivity. Unlike a wide beam, a narrow beam provides limited coverage and is unable to cover all the UEs in a cell. The narrow beam also cannot provide the maximum signal energy for each UE. For details, see [Figure 3-1](#). To address this issue, beam management has therefore been introduced to enable the gNodeB to separately manage beams for each type of channels and signals based on their respective characteristics, and select the optimal beams for UEs. This improves both the coverage performance of each type of channels and signals as well as user experience.

Figure 3-1 Comparison between wide and narrow beams



The NR system supports beam management in both high and low frequency bands. This document describes beam management only in high frequency bands. For details about beam management in low frequency bands, see *Beam Management*.

With beam management in high frequency bands, the gNodeB manages beams on millimeter wave (mmWave) to select the optimal beams for transmission and reception with respect to each type of channel and signal.

Beam management in high frequency bands involves broadcast beams and control and data beams, which are managed differently due to varying performance requirements. [Table 3-1](#) details the beam management methods.

Table 3-1 Beam management methods

Beam Classification	Applicable Channel and Signal	Management Method	Section
Broadcast beam	Physical broadcast channel (PBCH) and synchronization signal (SS)	The coverage of the cell PBCH and SS is managed to match cell coverage and UE distribution. Beams for multiple coverage scenarios are supported.	For details about beam management in the default scenario, see 4 mmWave Basic Beam Management . For details about beam management in other scenarios, see 5 mmWave 3D Coverage Pattern and 6 LampSite mmWave Scenario-based Beam .
Control and data beam	Physical downlink shared channel (PDSCH), physical uplink shared channel (PUSCH), physical uplink control channel (PUCCH), physical downlink control channel (PDCCH), channel state information-reference signal (CSI-RS), phase-tracking reference signal (PT-RS), and sounding reference signal (SRS)	Sufficient control and data beams are designed on demand to meet the coverage requirements of all UEs in a cell.	For details, see mmWave Control and Data Beam Management .

Beam Classification	Applicable Channel and Signal	Management Method	Section
Note: - Each PBCH, PDSCH, PUSCH, PUCCH, or PDCCH has its own demodulation reference signal (DMRS). The classification of DMRS beams is the same as beam classification of the corresponding channel. - For details about channels and signals, see <i>Channel Management</i>.			

 NOTE

PRACH beam management depends on SSB beam management. That is, the number and pattern of PRACH beams must be consistent with those of SSB beams.

4 mmWave Basic Beam Management

4.1 Principles

mmWave basic beam management is classified into mmWave broadcast beam management and mmWave control and data beam management.

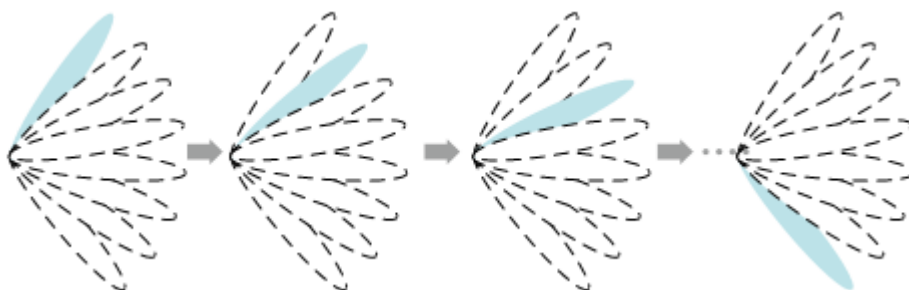
mmWave Broadcast Beam Management

mmWave broadcast beam management is specific to beams for the physical broadcast channel (PBCH) and synchronization signals (SS) in a cell. This extends the coverage area of the PBCH and SS and better matches the cell coverage scope and UE distribution.

The PBCH and SS share one Synchronization Signal and PBCH block (SSB) beam. **Figure 4-1** shows the SSB beams in an NR cell. One SSB beam is transmitted in a specific direction at a specific moment and then other SSB beams are transmitted in different directions at different moments in one SSB period (specified by ***NRDCell.SsbPeriod***) until the entire cell is covered.

A high-frequency UE requires sweeping for multiple beams on different panels and can sweep only one beam to measure all SSB beams of the base station within one SSB period. Therefore, the sweeping process takes a long time.

Figure 4-1 SSB beams



A UE receives signals from multiple beams and selects the beam with the best signal quality for synchronization messages and system information demodulation.

mmWave basic beam management applies only to broadcast beams in the default scenario, where the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter is set to **DEFAULT**.

mmWave basic beam management has certain requirements for RF modules. **Table 4-1** lists the specifications of the RF modules supporting mmWave basic beam management.

Table 4-1 RF modules supporting mmWave basic beam management and related specifications

RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a
HAAU5213 4T4R (28 GHz)	(87°, 18°)	-12° to +6°
HAAU5222 4T4R (28 GHz)	(79°, 16°)	-9° to +6°
HAAU5222 2T2R (28 GHz)	(79°, 17°)	-9° to +6°
HAAU5222 4T4R (26 GHz)	(85°, 15°)	-9° to +6°
HAAU5222 2T2R (26 GHz)	(85°, 17°)	-9° to +6°
HAAU5323 8T8R (28 GHz)	(84°, 17°)	-6° to +6°
HAAU5323 4T4R (28 GHz)	(84°, 13°)	-6° to +6°
HAAU5323 8T8R (26 GHz)	(84°, 19°)	-6° to +6°
HAAU5323 4T4R (26 GHz)	(84°, 15°)	-6° to +6°
HAAU5323 4T4R (26 GHz)	(102°, 12°)	-6° to +6°
pRRU5551JR 2T2R (26 GHz)	(360°, ±60°)	0°
HAAU5233w 4T4R (26 GHz)	(110°, 18°)	0° to 12°
HAAU5235w 4T4R (28 GHz)	(110°, 18°)	0° to 12°

RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a
a: The tilt can be adjusted by using the NRDUCellTrpMmwavBeam.Tilt parameter for HAAUs of 5900 series macro base stations. A negative value of this parameter specifies the physical beam tilt for upward adjustment, and a positive value of this parameter specifies the physical beam tilt for downward adjustment. For pRRUs of DBS5900 LampSite base stations, the tilt (NRDUCellTrpMmwavBeam.Tilt) must be set to 255 and cannot be adjusted; in this case, the default downtilt supported by the RF module is used.		

NOTE

- In the default scenario (with the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter set to **DEFAULT**), the azimuth (**NRDUCellTrpMmwavBeam.Azimuth**) must be set to **0** and cannot be adjusted.
- For the HAAU5323 (4T4R, 26 GHz):
 - With the 4:1 slot configuration (4_1_DDDSU), the horizontal 3 dB beamwidth and vertical 3 dB beamwidth take the values of 84° and 15°, respectively.
 - With the 2:3 slot configuration (2_3_DSUUU), the horizontal 3 dB beamwidth and vertical 3 dB beamwidth take the values of 102° and 12°, respectively.

The slot configuration can be configured using the **NRDUCell.SlotAssignment** parameter. For details, see *Standards Compliance*.

For 5900 series macro base stations, mmWave also supports the capacity, coverage, single-beam, and other scenarios to adapt to diversified requirements. For details, see [5 mmWave 3D Coverage Pattern](#).

For DBS5900 LampSite, mmWave also supports the indoor small multi-partition scenario to adapt to actual requirements. For details, see [6 LampSite mmWave Scenario-based Beam](#).

mmWave Control and Data Beam Management

mmWave control and data beam management uses CSI-RS beam sweeping to determine the optimal beams for each UE and implement real-time updates during UE movement, ensuring that each UE is served by the optimal beams.

mmWave control and data beam management involves downlink and uplink channels and signals.

- mmWave beam management for downlink channels and signals
Based on CSI-RS beam measurement, the gNodeB obtains the reference signal received power (RSRP) of downlink channels and signals, and selects a set of beams with the maximum RSRP (meaning the set of optimal beams) for the downlink channels and signals.
- mmWave beam management for uplink channels and signals
The optimal downlink beams for a UE are also used to receive uplink channels and signals (including the PUCCH, PUSCH, and SRS) from this UE.

4.2 Network Analysis

4.2.1 Benefits

Benefits of mmWave Broadcast Beam Management

mmWave broadcast beam management directs the transmit energy at target UEs by selecting the optimal beams for UEs. This increases demodulation signal-to-noise ratios (SNRs) of target UEs and improves the transmission success rate, and also extends the coverage area of the PBCH and SS, thereby increasing the cell coverage radius.

Benefits of mmWave Control and Data Beam Management

mmWave control and data beam management improves the beam directivity accuracy by selecting the optimal beams for UE, thereby increasing the demodulation SNRs of target UEs, improving the transmission success rate, and extending the coverage area of the control and data channels and signals.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

DBS5900 LampSite, which must be configured with the BBU5900 or BBU5910

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5213, HAAU5222, HAAU5323, HAAU5233w, HAAU5235w, pRRU5551JR (RHUB5873)

4.3.4 Cells

In the default scenario (with **NRDUCellTrpMmwavBeam.CoverageScenario** set to **DEFAULT**):

- The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 180, 188, 189, 192, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
- The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **186, 188, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
 - If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **186**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **1160**.
 - If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **188** or **65535**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **2410**.

4.3.5 Others

All cells in the same sector must have identical configurations of **NRDUCellTrpMmwavBeam.CoverageScenario** and **NRDUCellTrpMmwavBeam.Tilt**.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function takes effect by default and no parameters are involved in function activation. [Table 4-2](#) describes the parameters used for function optimization.

Table 4-2 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Coverage Scenario	NRDUCellTrpMmwav-Beam.CoverageScenario	Retain the default value. It is a high-risk parameter because changing the coverage scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Tilt	NRDUCellTrpMmwav-Beam.Tilt	Set this parameter based on the network plan. If this parameter is set to 255 , the default tilt 0° is used. It is a high-risk parameter because changing the cell tilt will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Changing from the coverage scenario of mmWave 3D coverage pattern to the default scenario (using 4 x 100 MHz as an example)
//Deactivating all cells in the related sector before modification
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Reconfiguring the CoverageScenario and Tilt parameters to identical values for all DU cells in the same sector
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=0, CoverageScenario=DEFAULT, Tilt=255;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT, Tilt=255;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=2, CoverageScenario=DEFAULT, Tilt=255;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=3, CoverageScenario=DEFAULT, Tilt=255;
//Activating all cells in the related sector after modification
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This function is enabled by default and does not require activation verification.

4.4.3 Network Monitoring

This function is a basic function that ensures basic service performance. It does not need to be monitored separately.

5 mmWave 3D Coverage Pattern

With mmWave 3D coverage pattern, the gNodeB supports mmWave beams for multiple coverage scenarios, to extend the coverage area of broadcast channels and synchronization signals in different networking scenarios and better match the cell coverage scope and UE distribution. Operators can select different beam adjustment solutions to meet the capacity or coverage requirements of target areas.

5.1 Principles

3GPP defines time-domain SSB patterns based on different subcarrier spacing (SCS) and frequencies. Given a pattern, cell capacity and coverage capability can be adjusted by changing the number of SSBs in the pattern. For example, reducing the number of SSBs will spare resources for PDSCH data transmission, thereby expanding system capacity; increasing the number of SSBs will allow more beams to be generated, thereby improving coverage capability. For details about the time-domain patterns of SSB transmissions, see section "PBCH Channel Management" in *Channel Management*.

Initial network deployment in high frequency bands is more oriented to hotspot capacity than that in low frequency bands and considers also coverage in certain areas. Given this, mmWave 3D coverage pattern supports the following scenarios by adjusting the number of SSBs in a pattern:

- Default scenario: This scenario is recommended if the target deployment area does not fit a specified application scenario, as shown in [Figure 5-1](#) and [Figure 5-2](#). For details, see [4 mmWave Basic Beam Management](#).

Figure 5-1 Default scenario with 4:1 slot configuration (4_1_DDDSU)



Figure 5-2 Default scenario with 2:3 slot configuration (2_3_DSUUU)



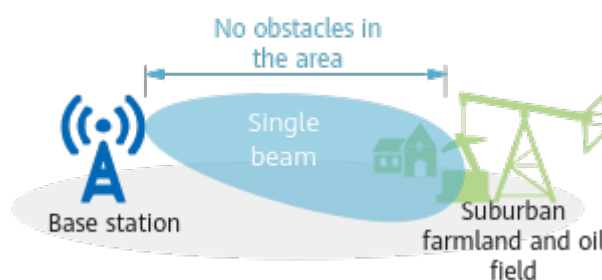
- Coverage scenario: This scenario is used to provide better coverage. It is recommended when (1) the coverage distance of base stations is long (with an inter-site distance of more than 500 meters) and (2) the downlink coverage is limited during the access in high-frequency deployment areas. See [Figure 5-3](#). In this release, this scenario applies only to fixed wireless access (FWA) services in SA networking.

Figure 5-3 Coverage scenario



- Wide-beam scenario: This scenario is recommended if (1) a base station provides LOS coverage of good signal quality for the target UE, which is far away from the base station, in high-frequency deployment areas, (2) the vertical coverage angle is narrow, and (3) the horizontal coverage angle is wide, as shown in [Figure 5-4](#). This scenario can be used to improve the downlink capacity.

Figure 5-4 Wide-beam scenario



- Capacity scenario: This scenario is recommended if (1) there are high capacity requirements due to densely distributed target UEs and (2) the coverage distance is short in high-frequency deployment areas.
 - In high-frequency deployment areas with 4:1 slot configuration (4_1_DDDSU), the capacity scenario is recommended if (1) there are high capacity requirements due to densely distributed target UEs and (2) the coverage distance is short (for example, in stadiums and small squares in front of business centers), as shown in [Figure 5-5](#).

Figure 5-5 Capacity scenario with 4:1 slot configuration (4_1_DDDSU)



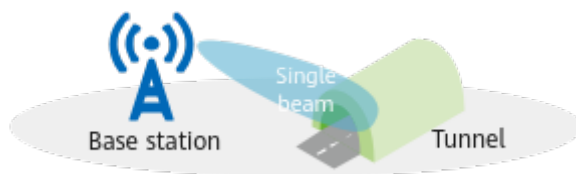
- In high-frequency deployment areas with 2:3 slot configuration (2_3_DSUUU), the capacity scenario is recommended if (1) there are high capacity requirements due to densely distributed target UEs and (2) the coverage scope is large (for example, in factories and ports), as shown in [Figure 5-6](#). Given that beams are discrete and discontinuous, the dents between adjacent beams will cause signal strength attenuation. In this context, this scenario can be used to narrow the dents between beams.

Figure 5-6 Capacity scenario with 2:3 slot configuration (2_3_DSUUU)



- Single-beam scenario: This scenario is used to improve experience. It is recommended when targeted UEs are mainly located at narrow areas (such as tunnels) in high-frequency deployment areas. See [Figure 5-7](#).

Figure 5-7 Single-beam scenario



- Customized beam scenario: There are three SSB beam patterns as shown in [Figure 5-8](#). Different SSB beam patterns can be selected based on the horizontal and vertical coverage ranges required by target UEs. This scenario is recommended to provide better performance in capacity when UEs are unevenly distributed with severe cell coverage overlapping in high-frequency deployment areas, such as stadiums, airport waiting lounges, and scenic spots, as shown in [Figure 5-9](#).

The total number of SSB beams that can be selected is specified by the **NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber** parameter. The antenna gains of SSB beam patterns 1, 2, and 3 are in descending order.

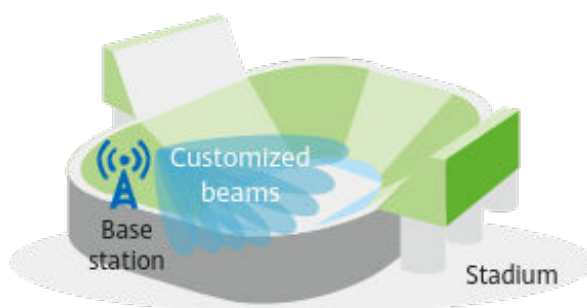
Figure 5-8 Beam IDs in different SSB beam patterns

SSB beam pattern 1: 64 SSB beams	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63

SSB beam pattern 2: 32 SSB beams	64	65	66	67	68	69	70	71
	72	73	74	75	76	77	78	79
	80	81	82	83	84	85	86	87
	88	89	90	91	92	93	94	95

SSB beam pattern 3: 16 SSB beams	96	97	98	99
	100	101	102	103
	104	105	106	107
	108	109	110	111

Figure 5-9 Customized beam scenario



In CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario), two adjacent cells may be served by different numbers of SSB beams. It is recommended that the **SSB Measurement Position Policy Switch** be turned on by selecting the **SSB_MEAS_POS_POLICY_SW** option of the **NRCellAlgoSwitch.MeasPolicySwitch** parameter. In this way, UEs can obtain accurate SSB beam information of a cell during cell handovers, thereby ensuring UE mobility. If the **SSB Measurement Position Policy Switch** is turned off, service drops may occur during cell handovers.

To further reduce intra-frequency interference to cell edge users (CEUs), the **CSI Beam Allocation Policy** (specified by the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter) is introduced to configure allocation policies according to scenarios.

- For stadium scenarios
 - Recommended configuration: It is recommended that customized beams be configured with SSB beam pattern 1. Under this configuration, it is recommended that the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter be set to **CSI_OPT_SEL_BEAM**. This mitigates the intra-frequency

interference to CEUs by reducing the number of CSI-RS beams, but slightly lowers the average downlink cell throughput.

- Not recommended configuration: Customized beams are configured with SSB beam pattern 2 or 3.
 - If the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter is set to **NOT_CONFIG** under this configuration, the number of CSI-RS beams is larger than that under the recommended configuration. Therefore, the intra-frequency interference to CEUs is higher than that under the recommended configuration.
 - If the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter is set to **CSI_OPT_SEL_BEAM** under this configuration, the average downlink cell throughput may decrease significantly and even service drops may occur because the CSI-RS beam coverage is smaller than the SSB beam coverage.
- In scenarios such as airport waiting lounges and scenic spots, the intra-frequency interference to CEUs is far less than that in stadium scenarios due to different user distributions.
 - If customized beams are configured with SSB beam pattern 2 or 3 and the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter is set to **CSI_OPT_SEL_BEAM**, the average downlink cell throughput may decrease significantly and even service drops may occur because the coverage range of CSI-RS beams is smaller than that of SSB beams. Therefore, it is recommended that the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter be set to **NOT_CONFIG** when customized beams are configured with SSB beam pattern 2 or 3.
 - If customized beams are configured with SSB beam pattern 1, the **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** parameter can be set to **CSI_OPT_SEL_BEAM** or **NOT_CONFIG**.

The gNodeB can select different coverage scenarios based on the setting of the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter. In each scenario, the tilt of an RF module in a cell can be adjusted flexibly and remotely. After the tilt adjustment, the cell coverage scope and shape will change to better cover the target area.

mmWave 3D coverage pattern has certain requirements for RF modules. [Table 5-1](#) lists the specifications of the RF modules supporting mmWave 3D coverage pattern.

Table 5-1 RF modules supporting mmWave 3D coverage pattern and related specifications

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
SCENARIO_101 (capacity scenario)	HAAU5213 4T4R (28 GHz)	(87°, 18°)	-12° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5222 4T4R (28 GHz)	(76°, 16°)	-9° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5222 2T2R (28 GHz)	(76°, 17°)	-9° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5222 4T4R (26 GHz)	(82°, 15°)	-9° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5222 2T2R (26 GHz)	(82°, 17°)	-9° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5323 8T8R (28 GHz)	(84°, 17°)	-6° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5323 4T4R (28 GHz)	(84°, 13°)	-6° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5323 8T8R (26 GHz)	(84°, 19°)	-6° to +6°	0°
SCENARIO_101 (capacity scenario)	HAAU5323 4T4R (26 GHz)	(84°, 15°)	-6° to +6°	0°

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
SCENARIO_101 (capacity scenario)	HAAU5233w 4T4R (26 GHz)	(110°, 18°)	0° to 12°	0°
SCENARIO_101 (capacity scenario)	HAAU5235w 4T4R (28 GHz)	(110°, 18°)	0° to 12°	0°
SCENARIO_102 (coverage scenario)	HAAU5323 8T8R (28 GHz)	(84°, 5°)	-6° to +6°	0°
SCENARIO_102 (coverage scenario)	HAAU5323 4T4R (28 GHz)	(84°, 5°)	-6° to +6°	0°
SCENARIO_102 (coverage scenario)	HAAU5323 8T8R (26 GHz)	(84°, 5°)	-6° to +6°	0°
SCENARIO_102 (coverage scenario)	HAAU5323 4T4R (26 GHz)	(84°, 5°)	-6° to +6°	0°
SCENARIO_102 (coverage scenario)	HAAU5233w 4T4R (26 GHz)	(110°, 12°)	0° to 12°	0°
SCENARIO_102 (coverage scenario)	HAAU5235w 4T4R (28 GHz)	(110°, 12°)	0° to 12°	0°
SCENARIO_103 (single-beam scenario)	HAAU5222 2T2R (28 GHz)	(9°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5222 4T4R (28 GHz)	(9°, 8°)	-6° to +6°	-10° to +10°

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
SCENARIO_103 (single-beam scenario)	HAAU5222 2T2R (26 GHz)	(9°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5222 4T4R (26 GHz)	(9°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5323 4T4R (28 GHz)	(7°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5323 8T8R (28 GHz)	(7°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5323 4T4R (26 GHz)	(7°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5323 8T8R (26 GHz)	(7°, 8°)	-6° to +6°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5233w 4T4R (26 GHz)	(9°, 8°)	0° to 12°	-10° to +10°
SCENARIO_103 (single-beam scenario)	HAAU5235w 4T4R (28 GHz)	(7°, 6°)	0° to 12°	-10° to +10°
SCENARIO_104 (wide-beam scenario)	HAAU5323 8T8R (28 GHz)	(84°, 9°)	-6° to +6°	0°
SCENARIO_104 (wide-beam scenario)	HAAU5323 4T4R (28 GHz)	(84°, 8°)	-6° to +6°	0°

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
SCENARIO_104 (wide-beam scenario)	HAAU5323 8T8R (26 GHz)	(84°, 9°)	-6° to +6°	0°
SCENARIO_104 (wide-beam scenario)	HAAU5323 4T4R (26 GHz)	(84°, 8°)	-6° to +6°	0°
SCENARIO_104 (wide-beam scenario)	HAAU5233w 4T4R (26 GHz)	(110°, 6°)	0° to 12°	0°
SCENARIO_104 (wide-beam scenario)	HAAU5235w 4T4R (28 GHz)	(100°, 6°)	0° to 12°	0°
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5323 4T4R (28 GHz)	- SSB beam pattern 1: (7°, 8°) for a single beam - SSB beam pattern 2: (14°, 8°) for a single beam - SSB beam pattern 3: (28°, 8°) for a single beam	0°	0°

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5323 8T8R (28 GHz)	- SSB beam pattern 1: (7°, 8°) for a single beam - SSB beam pattern 2: (14°, 8°) for a single beam - SSB beam pattern 3: (28°, 8°) for a single beam	0°	0°
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5323 4T4R (26 GHz)	- SSB beam pattern 1: (7°, 8°) for a single beam - SSB beam pattern 2: (14°, 8°) for a single beam - SSB beam pattern 3: (28°, 8°) for a single beam	0°	0°
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5323 8T8R (26 GHz)	- SSB beam pattern 1: (7°, 8°) for a single beam - SSB beam pattern 2: (14°, 8°) for a single beam - SSB beam pattern 3: (28°, 8°) for a single beam	0°	0°

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
CUSTOMIZE D_BEAM_SC ENARIO_10 6 (customized beam scenario)	HAAU5233w 4T4R (26 GHz)	- SSB beam pattern 1: (7°, 6°) for a single beam - SSB beam pattern 2: (14°, 6°) for a single beam - SSB beam pattern 3: (28°, 6°) for a single beam	0°	0°
CUSTOMIZE D_BEAM_SC ENARIO_10 6 (customized beam scenario)	HAAU5235w 4T4R (28 GHz)	- SSB beam pattern 1: (7°, 6°) for a single beam - SSB beam pattern 2: (14°, 6°) for a single beam - SSB beam pattern 3: (28°, 6°) for a single beam	0°	0°
a: The tilt can be adjusted using the NRDUCellTrpMmwavBeam.Tilt parameter. A negative value of this parameter specifies the physical beam tilt for upward adjustment, and a positive value of this parameter specifies the physical beam tilt for downward adjustment. b: The azimuth can be adjusted using the NRDUCellTrpMmwavBeam.Azimuth parameter. A negative value of this parameter specifies the physical beam azimuth for clockwise adjustment by using the pole as the center, and a positive value of this parameter specifies the physical beam azimuth for counterclockwise adjustment by using the pole as the center.				

NOTE

- In SCENARIO_101 (capacity scenario), SCENARIO_102 (coverage scenario), SCENARIO_104 (wide-beam scenario), and CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario), the azimuth cannot be adjusted because the horizontal coverage range of SSB beams has reached the upper limit.
- In CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario), SSB beams have a high degree of freedom in angle range vertically. Beams of different SSB beam patterns can be selected for coverage. In this case, tilt adjustment is not supported.
- When the tilt of an SSB beam is adjusted by using **NRDUCellTrpMmwavBeam.Tilt** or the azimuth of an SSB beam is adjusted by using **NRDUCellTrpMmwavBeam.Azimuth**, the tilt or azimuth of the CSI-RS beam is also adjusted.

The **NRDUCellTrpMmwavBeam.MaxSsbPwrOffset** parameter can be reconfigured to change the maximum power offset for SSB transmission, thereby improving cell edge coverage. For details, see *Power Control*.

Example of Coverage Scenario Selection

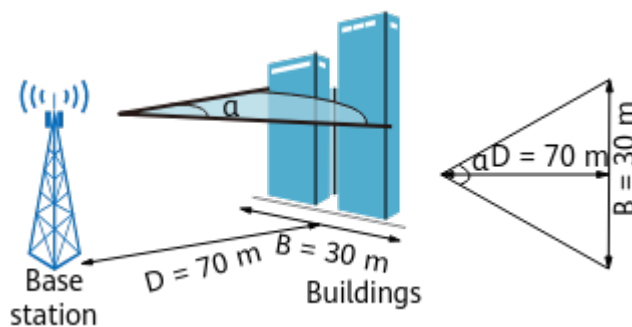
Operators of live networks need to determine coverage scenarios based on factors such as site height, inter-site distance, UE distribution, coverage scope, and RF module capabilities.

The following is an example of how to perform calculation and select a coverage scenario. Assume that the horizontal 3 dB beamwidth is indicated by α in [Figure 5-10](#) and the vertical 3 dB beamwidth is indicated by β in [Figure 5-11](#). They together determine the coverage scenario. In the figures, B indicates the total width of buildings, $D/D1/D2$ indicates the distance between the base station and the buildings, H indicates the height of the base station, and $h1/h2$ indicates the height of the buildings.

Step 1 Calculate the horizontal 3 dB beamwidth.

If B is 30 m and D is 70 m, then α is 24° , as shown in [Figure 5-10](#).

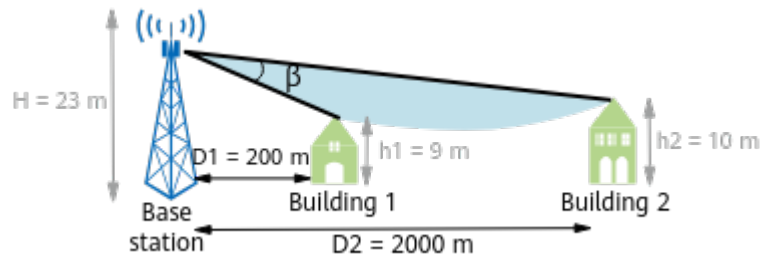
Figure 5-10 Calculating the horizontal 3 dB beamwidth



Step 2 Calculate the vertical 3 dB beamwidth.

If H is 23 m, $D1$ is 200 m, $D2$ is 2000 m, $h1$ is 9 m, and $h2$ is 10 m, then β is 4° , as shown in [Figure 5-11](#).

Figure 5-11 Calculating the vertical 3 dB beamwidth



Step 3 Based on the angles in [Step 1](#) and [Step 2](#), the appropriate scenarios are SCENARIO_101, SCENARIO_102, SCENARIO_104, and CUSTOMIZED_BEAM_SCENARIO_106 according to [Table 5-1](#).

Step 4 For CUSTOMIZED_BEAM_SCENARIO_106, a specific SSB beam ID set needs to be determined based on the calculation results in steps 1 and 2 and the application scenario. [Figure 5-13](#) shows the IDs of beams under different SSB beam patterns.

- In stadium scenarios, only beams of SSB beam pattern 1 can be selected.
 - Angle θ between a beam and the horizontal normal line:
 - In the target coverage area A, the angle between a beam and the horizontal normal line refers to θ_1 between CG and DG. D is the center of target area A; F is the center of the area to which the RF module points; C is the intersection point where F is perpendicular to the horizontal extension line of D; G is the center of the RF module. For details, see [Figure 5-12](#).
Assume that an RF module needs to cover target area A and D is located on the right of F. In this scenario, the beam's horizontal normal line will move rightwards by an angle of θ_1 from the position of the beam's horizontal normal line with θ being 0° as shown in [Figure 5-13](#). [Figure 5-14](#) shows the position of the beam's horizontal normal line with θ_1 being 9° .
 - In the target coverage area B, the angle between a beam and the horizontal normal line refers to θ_2 between EG and KG. E is the center of target area B; F is the center of the area to which the RF module points; K is the intersection point where F is perpendicular to the horizontal extension line of E; G is the center of the RF module. For details, see [Figure 5-12](#).
Assume that an RF module needs to cover target area B and E is located on the left of F. In this scenario, the beam's horizontal normal line will move leftwards by an angle of θ_2 from the position of the beam's horizontal normal line with θ being 0° as shown in [Figure 5-13](#). [Figure 5-14](#) shows the position of the beam's horizontal normal line with θ_2 being 9° .
 - Number of horizontal beams: $\text{Roundup}(((\alpha/2)/\text{Horizontal 3 dB beamwidth per beam in SSB beam pattern 1}) \times 2)$.
 - If the "angle between a beam and the horizontal normal line" (θ) is 0° and the calculated number of horizontal beams is 2.67 (rounded up to 3), three beams of SSB beam pattern 1 are selected on the left and right sides of the beam's horizontal normal line with θ being 0° separately. The horizontal beam IDs are 5, 6, 7, 8, 9, and 10.

- If the "angle between a beam and the horizontal normal line" (θ) is not 0° and the angle is divisible by 7:
 - Assume that the horizontal coverage center is on the left of the horizontal center of the area to which the RF module points and θ_1 is 7° . For details, see [Figure 5-12](#). If the calculated number of horizontal beams is 2.67 (rounded up to 3), three beams of SSB beam pattern 1 are selected on the left and right sides of the beam's horizontal normal line with θ_1 being 7° separately. The horizontal beam IDs are 4, 5, 6, 7, 8, and 9.
 - Assume that the horizontal coverage center is on the right of the horizontal center of the area to which the RF module points and θ_2 is 7° . For details, see [Figure 5-12](#). If the calculated number of horizontal beams is 2.67 (rounded up to 3), three beams of SSB beam pattern 1 are selected on the left and right sides of the beam's horizontal normal line with θ_2 being 7° separately. The horizontal beam IDs are 6, 7, 8, 9, 10, and 11.
- If the "angle between a beam and the horizontal normal line" (θ) is not 0° and the angle is indivisible by 7:
 - Assume that the horizontal coverage center is on the left of the horizontal center of the area to which the RF module points and θ_1 is 9° . For details, see [Figure 5-12](#). If the calculated number of horizontal beams is 2.67 (rounded up to 3), beam 6 where the beam's horizontal normal line is located is selected first, and then three beams of SSB beam pattern 1 are selected on the left and right sides of beam 6 separately. The horizontal beam IDs are 3, 4, 5, 6, 7, 8, and 9.
 - Assume that the horizontal coverage center is on the right of the horizontal center of the area to which the RF module points and θ_2 is 9° . For details, see [Figure 5-12](#). If the calculated number of horizontal beams is 2.67 (rounded up to 3), beam 9 where the beam's horizontal normal line is located is selected first, and then three beams of SSB beam pattern 1 are selected on the left and right sides of beam 9 separately. The horizontal beam IDs are 6, 7, 8, 9, 10, 11, and 12.
- Angle γ between a beam and the vertical normal line:
 - In the target coverage area A, the angle between a beam and the vertical normal line refers to γ_1 between CG and FG. D is the center of target area A; F is the center of the area to which the RF module points; C is the intersection point where F is perpendicular to the horizontal extension line of D; G is the center of the RF module. For details, see [Figure 5-12](#).
 - In the target coverage area B, the angle between a beam and the vertical normal line refers to γ_2 between KG and FG. E is the center of target area B; F is the center of the area to which the RF module points; K is the intersection point where F is perpendicular to the horizontal extension line of E; G is the center of the RF module. For details, see [Figure 5-12](#)
- Number of vertical beams: Roundup(β /Vertical 3 dB beamwidth per beam in SSB beam pattern 1).

- If the "angle between a beam and the vertical normal line" (γ) is 0° and the calculated number of vertical beams is rounded up to 1, the beams at the layer above the vertical normal line are preferentially selected, that is, the beams at the second layer. If the roundup value is 2, the beams at the second and third layers are selected. If the roundup value is 3, the beams at the second, third, and fourth layers are selected. If the roundup value is 4, the beams at four layers are selected.
- If the "angle between a beam and the vertical normal line" (γ) is not 0° and the angle is divisible by 7:
 - Assume that the center of target coverage area is above (from the perspective of point G, point C or K is above point F) the center of the area to which the RF module points and γ_1 is 7° . For details, see [Figure 5-15](#). In this scenario, the beam's vertical normal line moves 7° upwards from the position of the beam's vertical normal line with γ being 0° as shown in [Figure 5-13](#), and is located at the boundary between the first and second beam layers. If the calculated number of vertical beams is rounded up to 1, the beams at the layer below the vertical normal line are preferentially selected, that is, the beams at the second layer. If the roundup value is 2, the beams at the second and third layers are selected. If the roundup value is 3, the beams at the first, second, and third layers are selected. If the roundup value is 4, the beams at four layers are selected.
 - Assume that the center of target coverage area is below (from the perspective of point G, point C or K is below point F) the center of the area to which the RF module points and γ_2 is 7° . In this scenario, the beam's vertical normal line moves 7° downwards from the position of the beam's vertical normal line with γ being 0° as shown in [Figure 5-13](#), and is located at the boundary between the third and fourth layers. If the calculated number of vertical beams is rounded up to 1, the beams at the layer above the vertical normal line are preferentially selected, that is, the beams at the third layer. If the roundup value is 2, the beams at the third and fourth layers are selected. If the roundup value is 3, the beams at the second, third, and fourth layers are selected. If the roundup value is 4, the beams at four layers are selected.
- If the "angle between a beam and the vertical normal line" (γ) is not 0° and the angle is indivisible by 7:
 - Assume that the center of target coverage area is above (from the perspective of point G, point C or K is above point F) the center of the area to which the RF module points and γ_1 is 3° . For details, see [Figure 5-15](#). In this scenario, the beam's vertical normal line moves 3° upwards from the position of the beam's vertical normal line with γ being 0° as shown in [Figure 5-13](#), and is located at the second layer. If the calculated number of vertical beams is rounded up to 1, the beams at the second layer are selected. If the roundup value is 2, the beams at the second and third layers are selected. If the roundup value is 3, the

beams at the first, second, and third layers are selected. If the roundup value is 4, the beams at four layers are selected.

- Assume that the center of target coverage area is below (from the perspective of point G, point C or K is below point F) the center of the area to which the RF module points and γ_2 is 3° . In this scenario, the beam's vertical normal line moves 3° downwards from the position of the beam's vertical normal line with γ being 0° as shown in [Figure 5-13](#), and is located at the third layer. If the calculated number of vertical beams is rounded up to 1, the beams at the third layer are selected. If the roundup value is 2, the beams at the third and fourth layers are selected. If the roundup value is 3, the beams at the second, third, and fourth layers are selected. If the roundup value is 4, the beams at four layers are selected.

The **Customized SSB Beam Number** (specified by the **NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber** parameter) cannot exceed 16. If over 16 beams are selected after using the preceding method, calculate the result of $\text{Rounddown}(((\alpha/2)/\text{Horizontal 3 dB beamwidth per beam in SSB beam pattern 1}) \times 2)$ and then check whether the total number of beams exceeds 16. If the total number still exceeds 16, calculate the result of $\text{"Rounddown}(((\alpha/2)/\text{Horizontal 3 dB beamwidth per beam in SSB beam pattern 1}) \times 2) - 1"$ and then check the total number of beams. If the total number still exceeds 16, calculate the result of $\text{Rounddown}(\beta/\text{Vertical 3 dB beamwidth per beam in SSB beam pattern 1})$, and then check the total number of beams again. If the total number still exceeds 16, remove one or two more layers of vertical beams near the normal line until the number of beams does not exceed 16.

- For non-stadium scenarios (such as airport waiting lounges and scenic spots):
The methods of calculating (1) the angle between a beam and the horizontal normal line (θ), (2) number of horizontal beams, (3) angle between a beam and the vertical normal line (γ), and (4) number of vertical beams are the same as those in the preceding stadium scenario. If over 16 beams are selected after using the preceding methods, beams are selected as follows:
Beams of SSB beam pattern 1 are preferentially replaced with those of SSB beam pattern 2 with the same coverage. If the number of beams still exceeds 16 after the replacement, beams of SSB beam pattern 1 are replaced with beams of SSB beam pattern 3 with the same coverage. For example, if beams 0 to 47 of SSB beam pattern 1 are selected, they can be replaced with beams 64 to 87 of SSB beam pattern 2. However, the number of beams still exceeds 16, the beams of SSB beam pattern 1 need to be replaced with beams 96 to 107 of SSB beam pattern 3. In this way, the number of beams is 12, which is less than 16 and meets the requirements.

The HAAU5323 is used as an example. Assume that the angles between a beam and the horizontal (θ) and vertical (γ) normal lines are both 0° . Based on the calculation results in steps 1 and 2, regardless of stadium and non-stadium scenarios, the horizontal beams 4, 5, 6, 7, 8, 9, 10, and 11 are selected and there is one vertical beam which indicates that only one layer of vertical beams is required. In this way, the selected beam IDs are 4, 5, 6, 7, 8, 9, 10, and 11.

Figure 5-12 Angles between a beam and the normal lines

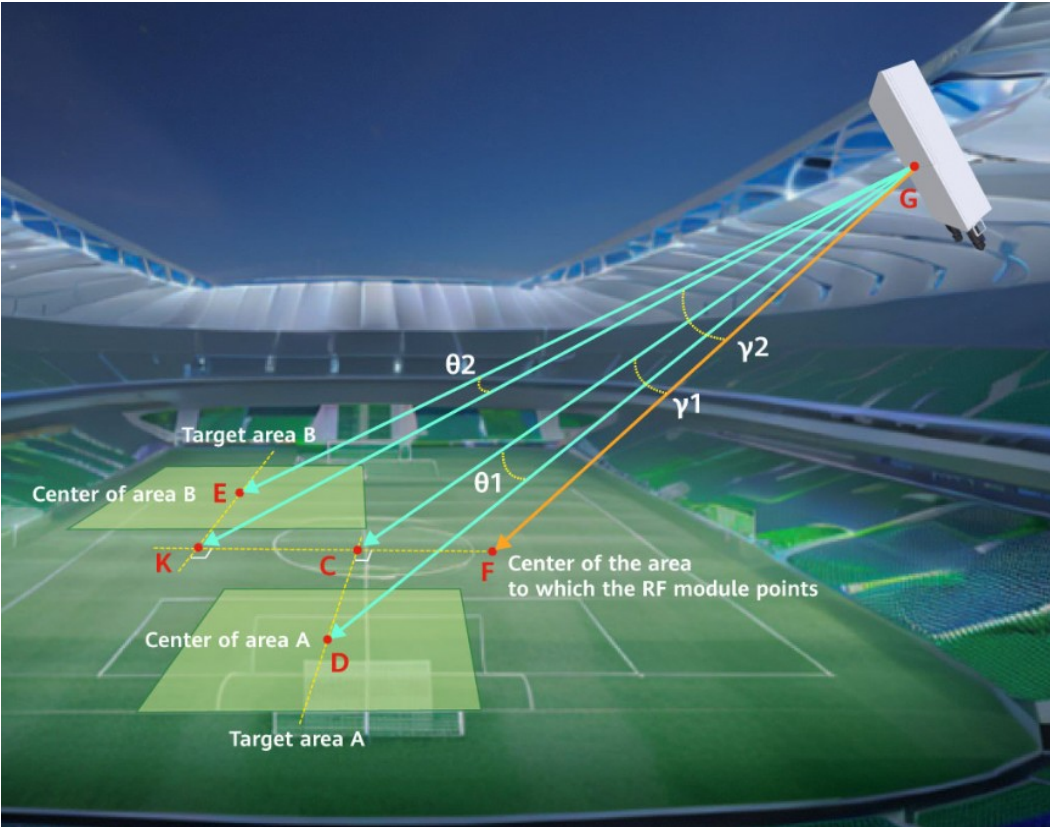


Figure 5-13 Beam IDs in different SSB beam patterns

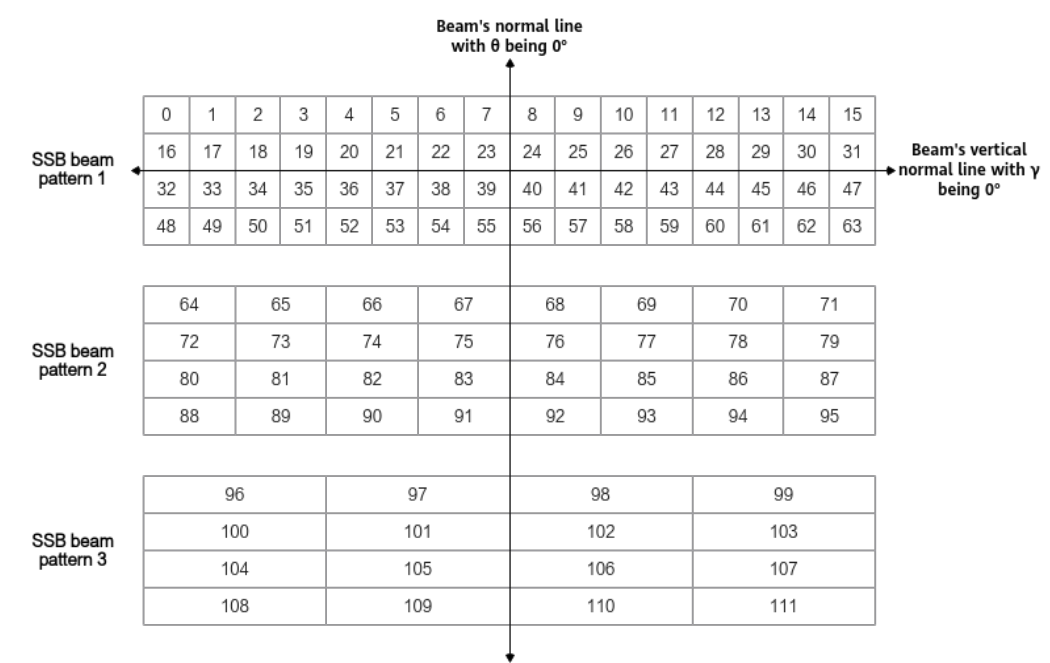


Figure 5-14 Positions of a beam's horizontal normal lines with different angles

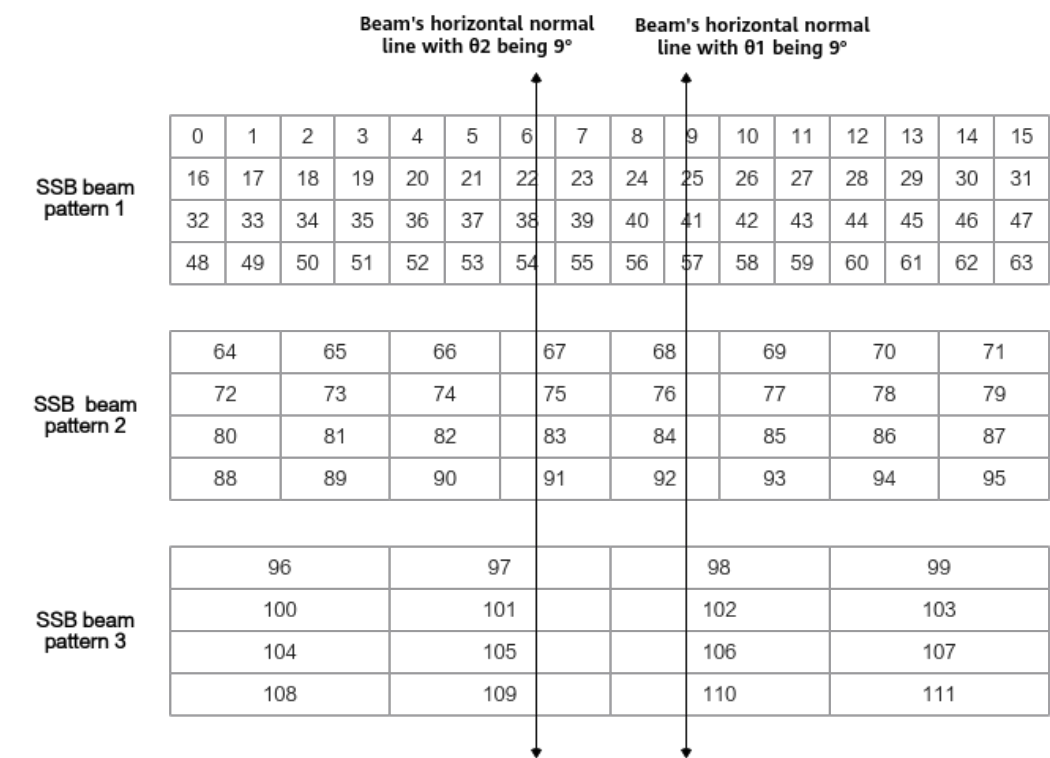
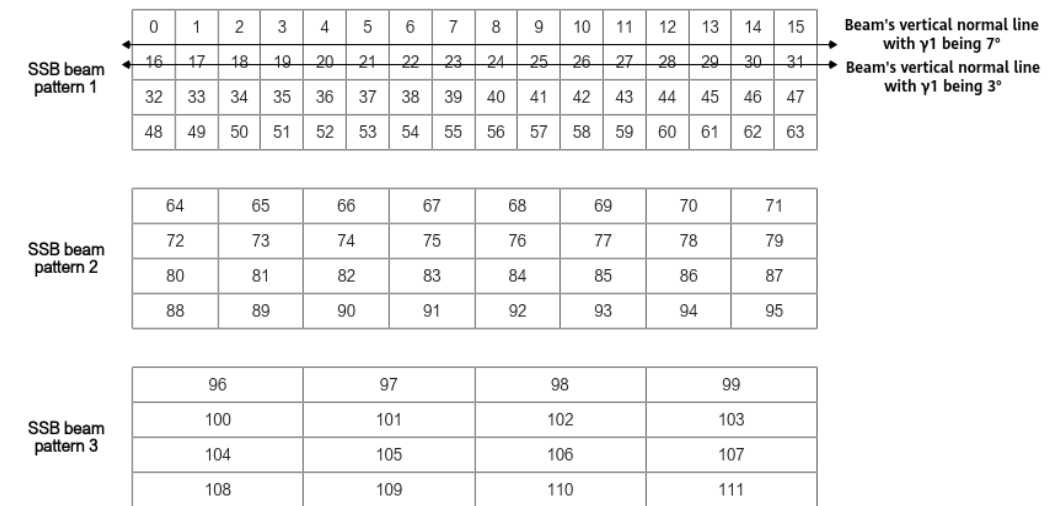


Figure 5-15 Positions of a beam's vertical normal lines with different angles



----End

5.2 Network Analysis

5.2.1 Benefits

3D coverage pattern allows scenario selection based on network conditions and UE distribution to ensure maximum coverage or capacity gains.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

After the beam scenario changes, coverage scopes of SSB and CSI-RS beams in a cell may change. As a result, the number of UEs in the cell, UE throughput, and other KPIs related to access, handovers, and service drops change.

In stadiums and other similar venues, the average downlink cell throughput slightly decreases if (1) **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy** is set to **CSI_OPT_SEL_BEAM**, (2) **NRDUCellTrpMmwavBeam.CoverageScenario** is set to **CUSTOMIZED_BEAM_SCENARIO_106**(Customized Beam Scenario), and (3) all the values of **NRDUCellTrpMmwavBeam.CustomizedSsbBeam0Id** to **NRDUCellTrpMmwavBeam.CustomizedSsbBeam15Id** are less than or equal to 63.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030201	mmWave 3D Coverage Pattern	NR0SMMW3DCP0	per Cell
For the license control item NR0SMMW3DCP0: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the NRDUCellTrpMmwavBeam.CoverageScenario parameter value cannot be changed to a value other than DEFAULT due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	CSI-RS for BM in SSB slots	SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	- The wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario)) and the customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario)) can be configured only when CSI-RS for BM in SSB slots is enabled by selecting the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter. - In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.

RAT	Function Name	Function Switch	Reference	Description
				<i>CoverageScenario</i> parameter set to SCENARIO_104(Wide Beam Scenario)), the gains of CSI-RS for BM in SSB slots decrease because the number of slots occupied by SSB beams decreases and resources for CSI-RS for BM in SSB slots decrease.
High-frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	The customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario)) can be configured only after Flexible Dense Beam is enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Tilt	NRDUCellTrpMmwavBeam.Tilt	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the customized beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario)), the tilt can only be set to 255 .
High-frequency TDD	Azimuth	NRDUCellTrpMmwavBeam.Azimuth	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the customized beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario)), the azimuth can only be set to 0 .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmWavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	Only in the customized beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmWavBeam.CoverageScenario parameter set to CUSTOMIZED_BEAM_SCENARIO_106 (Customized Beam Scenario)), the NRDUCellTrpMmWavBeam.CSIBeamAllocationPolicy parameter can be set to CSI_OPT_SEL_BEAM .

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Narrow-beam probing	NARROW_BEAM_PROBE_SW option of the NRDUCellIABFullChannelCoverageEnhSwitch parameter	<i>Random Access Control</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmWavBeam.CoverageScenario parameter set to SCENARIO_103 (Single Beam Scenario)), narrow-beam probing cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH receive channel adaptation	NRDUCellPucch.PucchRxAllocMode set to ADAPTIVE	<i>Intelligent Aggregate Scheduling (High-Frequency TDD)</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), PUCCH receive channel adaptation cannot be enabled.
High-frequency TDD	UE-specific TRS resource management	USER_TRS_BEAM_SW option of the NRDUCellCsiRs.CsiSwitch parameter	<i>Channel Management</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)) and wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_104(Wide Beam Scenario)), UE-specific TRS resource management cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	TRS Beam Type	NRDUCellCsirs.TrsBeamType	None	In the following scenarios, the TRS beam type (NRDUCellCsirs.TrsBeamType parameter) can only be set to NARROW_BEAM, rather than WIDE_BEAM. • Single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)) • Customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario))

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Beam-based power control	WRC19_EIRP_CONTROL_SW option of the NRDUCellsmartPwrLock.SmartPowerLockSw parameter	None	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), beam-based power control cannot be enabled.
High-frequency TDD	Grid-specific power control in the smart power lock	AVERAGE_3DEIRP_CONTROL_SW option of the NRDUCellsmartPwrLock.SmartPowerLockSw parameter	None	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), grid-specific power control in the smart power lock cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Adaptive number of downlink additional DMRS positions	NRDUCellPdsch.DLAdditionalDmrsPositions set to POS_ADAPTIVE	<i>Downlink Scheduling</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwaveBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), adaptive number of downlink additional DMRS positions cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Full-buffer UE-specific CSI-RS for BM	USER_BM_C SIRS_SW option of the NRDUCellCsiRs.CsiSwitch parameter	<i>Channel Management</i>	Full-buffer UE-specific CSI-RS for BM cannot be enabled in the following scenarios: • Single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)) • Wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario)) • The NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy parameter is set to CSI_OPT_SEL_BEAM.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink RMSI payload reduction	RMSI_PAYLOAD_REDUCE_SW option of the NRDUCellDL Sch.DLSchAlgorithmSwitch parameter	<i>Downlink Scheduling</i>	In the wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwaveBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario)), downlink RMSI payload reduction cannot be enabled.
High-frequency TDD	Intelligent beam selection	INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgorithmSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwaveBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), intelligent beam selection cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink Msg4 payload reduction	MSG4_PAYLOAD_REDUCTION_SW option of the NRDUCellDL-Sch.DischAlgoSwitch parameter	<i>Downlink Scheduling</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwaveBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), if the NRDUCellPucch.PucchResrcRednMthd parameter is set to MASS_FLOW , downlink Msg4 payload reduction cannot be enabled.
High-frequency TDD	Optimal selection of narrow beams for access channels	ACC_CHN_OPT_NARROW_BEAM_SELECTION_SW option of the NRDUCellTrpMmwaveBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwaveBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario)), optimal selection of narrow beams for access channels cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Antenna large physical downtilt mode	ANT_LARGE_PHY_TILT_MODE_SW option of the RRUEXTINFO.RRUOPTLFEATSW parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	The antenna large physical downtilt mode cannot be enabled in the following scenarios: • Coverage scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_102(Coverage Scenario)) • Single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)) • Wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario))

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-FR CA does not take effect when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103(Single Beam Scenario) .
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	mmWave PUSCH MU-MIMO cannot be enabled in the following scenarios: - Single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)) - Customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario))

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	High Speed Cell Handover Protect Timer	NRCellMobilityConfig.HsCellHoProtectTimer	None	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwaveBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), it is recommended that the High Speed Cell Handover Protect Timer parameter (NRCellMobilityConfig.HsCellHoProtectTimer) be set to 20 during deployment in tunnels.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	• In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), the switch for Flexible Dense Beam can be turned on, but the function does not take effect. • In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_104(Wide Beam Scenario)), the gains of Flexible Dense Beam may decrease because coverage scopes of SSB and CSI-RS beams may change.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Sparsified RMSI scheduling policy	NRDUCellDL Sch.RmsiSparsifiedSchPolicy set to a value other than OFF	<i>Downlink Scheduling</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmWaveBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), the sparsified RMSI scheduling policy function does not take effect.
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellALgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmWaveBeam.CoverageScenario parameter set to SCENARIO_104(Wide Beam Scenario)), the gains of mmWave PUSCH MU-MIMO may decrease because coverage scopes of SSB and CSI-RS beams may change.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Antenna large physical downtilt mode	ANT_LARGE_PHY_TILT_MODE_SW option of the RRUEXTINFO.RRUOPTLFEATSW parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	After the antenna large physical downtilt mode is enabled, the vertical 3 dB beamwidth is expanded in the following mmWave 3D coverage pattern scenarios: • Default scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to DEFAULT(Default Scenario)) • Capacity scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_101(Capacity Scenario)) • Customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario))

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Inter-FR CA coverage enhancement	PUCCH_GRO UP_ADAPT_S W option of the NRDUCellCa rrMgmt.Inte rFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Inter-FR CA coverage enhancement is not recommended when the NRDUCellTrpMm wavBeam.Covera geScenario parameter is set to SCENARIO_104(Wide Beam Scenario) . Inter-FR CA coverage enhancement is used in mmWave eMBB scenarios to address limited uplink coverage caused by limited UE transmit power in high frequency bands. It is recommended that the NRDUCellTrpMm wavBeam.Covera geScenario parameter be set to SCENARIO_104(Wide Beam Scenario) when the high-frequency LOS coverage is not limited, in which case the uplink coverage of high frequency bands is not limited. If inter-FR CA coverage enhancement takes effect unexpectedly in this case, the UE

RAT	Function Name	Function Switch	Reference	Description
				throughput decreases.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Capacity scenarios: HAAU5213, HAAU5222, HAAU5323, HAAU5233w, HAAU5235w. The HAAU5213 only supports the capacity scenario with the 4:1 slot configuration.

Single-beam scenarios: HAAU5222, HAAU5323, HAAU5233w, HAAU5235w

Other scenarios: HAAU5323, HAAU5233w, HAAU5235w

5.3.4 Cells

- For SCENARIO_101 (capacity scenario):
 - The capacity scenario (SCENARIO_101) of mmWave 3D coverage pattern is supported only when **NRDUCell.SlotAssignment** is set to **4_1_DDDSU** or **2_3_DSUUU** for the cell.
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 176, 180, 188, 189, 192**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **186, 188**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).

- If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **186**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **1160**.
- If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **188** or **65535**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **2410**.
- If the cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**, the SSB subcarrier spacing (**NRDUCell.SsbSubcarrierSpacing**) can only be set to **120KHZ**.
- For SCENARIO_102 (coverage scenario):
 - The coverage scenario (SCENARIO_102) of mmWave 3D coverage pattern is supported only when **NRDUCell.SlotAssignment** is set to **4_1_DDDSU** for the cell.
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 180, 188, 189, 192**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
- For SCENARIO_103 (single-beam scenario):
 - The single-beam scenario (SCENARIO_103) of mmWave 3D coverage pattern is supported only when **NRDUCell.SlotAssignment** is set to **4_1_DDDSU** for the cell.
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 176, 180, 188, 189, 192**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
- For SCENARIO_104 (wide-beam scenario):
 - The wide-beam scenario (SCENARIO_104) of mmWave 3D coverage pattern is supported only when **NRDUCell.SlotAssignment** is set to **4_1_DDDSU** for the cell.
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 180, 188, 189, 192**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
 - The **NRDUCellTrpMmwavBeam.CoverageScenario** parameter can only be set to **SCENARIO_104(Wide Beam Scenario)** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell

radius (**NRDUCell.CellRadius**) is set to a value in the range from **5001** to **10000**.

- CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario):
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **186, 188, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
 - If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **186**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **1160**.
 - If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **188** or **65535**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **2410**.
 - The number of beams in the customized SSB beam scenario (**NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber**) can only be set to **1** or **2** when both of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 180, 188, 189, 192, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**). (4) The number of customized SSB beams (**NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber**) is greater than **8**.
 - The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 176, 180, 188, 189, 192, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**). (4) The number of customized SSB beams (**NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber**) is less than or equal to **8**.
 - The SSB subcarrier spacing (**NRDUCell.SsbSubcarrierSpacing**) cannot be set to **240KHZ**.
 - The transmit and receive mode (**NRDUCellTrp.TxRxMode**) cannot be set to **8T8R** for a cell when the CSI beam allocation policy (**NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy**) is set to **CSI_OPT_SEL_BEAM**.

5.3.5 Others

- All activated cells in the same sector must have identical configurations of **NRDUCellTrpMmwavBeam.CoverageScenario**, **NRDUCellTrpMmwavBeam.Tilt**, and **NRDUCellTrpMmwavBeam.Azimuth**.
- In the CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario), all activated cells in a sector must share the same settings of the following parameters: **NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber**, **NRDUCellTrpMmwavBeam.CustomizedSsbBeam0Id** to **NRDUCellTrpMmwavBeam.CustomizedSsbBeam15Id**, and **NRDUCellTrpMmwavBeam.CSIBeamAllocationPolicy**. If the **NRDUCellTrpMmwavBeam.CustomizedSsbBeamNumber** parameter is set to a value other than 1, all the values of **NRDUCellTrpMmwavBeam.CustomizedSsbBeam0Id** to **NRDUCellTrpMmwavBeam.CustomizedSsbBeam15Id** must be unique.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 5-2 Required parameters

Parameter Name	Parameter ID	Option	Setting Notes
Coverage Scenario	NRDUCellTrpMmwavBeam.CoverageScenario	N/A	Before setting this parameter, confirm the coverage type of the target scenario. Without confirmation, it is good practice to select the default scenario. It is a high-risk parameter because changing the coverage scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam Number	NRDUCellTrpMmwav-Beam.CustomizedSsbBeamNumber	N/A	Before the configuration, determine the number of beams for coverage based on the target coverage scenario. If this parameter is set to a value other than 1, the values of Customized SSB Beam0 ID to Customized SSB Beam15 ID (specified by NRDUCellTrpMmwav-Beam.CustomizedSsbBeam0Id to NRDUCellTrpMmwav-Beam.CustomizedSsbBeam15Id) must be different. It is a high-risk parameter because changing the number of beams in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Measurement Policy Switch	NRCelAlgoSwitch.MeasPolicySwitch	SSB_MEAS_POS_POLICY_SW	It is recommended that this option be selected when the NRDUCellTrpMmwav-Beam.CoverageScenario parameter is set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario).

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam0 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam0Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam1 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam1Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam2 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam2Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam3 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam3Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam4 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam4Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam5 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam5Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam6 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam6Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam7 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam7Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam8 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam8Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam9 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam9Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam10 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam10Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam11 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam11Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam12 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam12Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam13 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam13Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
Customized SSB Beam14 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam14Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Customized SSB Beam15 ID	NRDUCellTrpMmwav Beam.CustomizedSsb Beam15Id	N/A	Before the configuration, determine the required beam IDs in the customized SSB beam scenario based on the target coverage scenario and the number of beams in the customized SSB beam scenario. It is a high-risk parameter because changing the beam ID in the customized SSB beam scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

Parameter Name	Parameter ID	Option	Setting Notes
CSI Beam Allocation Policy	NRDUCellTrpMmwaveBeam.CSIBeamAllocationPolicy	N/A	Before the configuration, determine the CSI beam allocation policy based on the configurations of customized beam scenarios. For details, see the related descriptions about customized beam scenario in 5.1 Principles .
Tilt	NRDUCellTrpMmwaveBeam.Tilt	N/A	Set this parameter based on the network plan. If this parameter retains the default value of 255 , the default tilt of 0° is used. It is a high-risk parameter because changing the cell tilt will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Azimuth	NRDUCellTrpMmwaveBeam.Azimuth	N/A	Set this parameter based on the network plan. The default value 0 is used. It is a high-risk parameter because changing the cell azimuth will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//The following uses 4 x 100 MHz as an example.
//Deactivating all cells in the related sector before modification
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Applying the same configuration for the CoverageScenario, Tilt, and Azimuth parameters across all DU
cells in the same sector when configuring the capacity, coverage, single-beam, and wide-beam scenarios
(The following uses the capacity scenario as an example.)
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0, CoverageScenario=SCENARIO_101, Tilt=255,
Azimuth=0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=1, CoverageScenario=SCENARIO_101, Tilt=255,
Azimuth=0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=2, CoverageScenario=SCENARIO_101, Tilt=255,
Azimuth=0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=3, CoverageScenario=SCENARIO_101, Tilt=255,
Azimuth=0;
//Applying the same configuration for the CoverageScenario, Tilt, and Azimuth parameters across all DU
cells in the same sector and setting the number of customized SSB beams and the corresponding beam IDs
when configuring the customized beam scenario (The following uses a stadium scenario with four SSB
beams as an example.)
MOD NRCELLALGOSWITCH: NrCellId=0, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=1, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=2, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=3, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-1;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0,
CoverageScenario=CUSTOMIZED_BEAM_SCENARIO_106, Tilt=255, Azimuth=0,
CustomizedSsbBeamNumber=4, CustomizedSsbBeam0Id=1, CustomizedSsbBeam1Id=2,
CustomizedSsbBeam2Id=3, CustomizedSsbBeam3Id=4, CSIBeamAllocationPolicy=CSI_OPT_SEL_BEAM;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=1,
CoverageScenario=CUSTOMIZED_BEAM_SCENARIO_106, Tilt=255, Azimuth=0,
CustomizedSsbBeamNumber=4, CustomizedSsbBeam0Id=1, CustomizedSsbBeam1Id=2,
CustomizedSsbBeam2Id=3, CustomizedSsbBeam3Id=4, CSIBeamAllocationPolicy=CSI_OPT_SEL_BEAM;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=2,
CoverageScenario=CUSTOMIZED_BEAM_SCENARIO_106, Tilt=255, Azimuth=0,
CustomizedSsbBeamNumber=4, CustomizedSsbBeam0Id=1, CustomizedSsbBeam1Id=2,
CustomizedSsbBeam2Id=3, CustomizedSsbBeam3Id=4, CSIBeamAllocationPolicy=CSI_OPT_SEL_BEAM;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=3,
CoverageScenario=CUSTOMIZED_BEAM_SCENARIO_106, Tilt=255, Azimuth=0,
CustomizedSsbBeamNumber=4, CustomizedSsbBeam0Id=1, CustomizedSsbBeam1Id=2,
CustomizedSsbBeam2Id=3, CustomizedSsbBeam3Id=4, CSIBeamAllocationPolicy=CSI_OPT_SEL_BEAM;
//Activating all cells in the related sector after modification
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
```

Optimization Command Examples

None

Deactivation Command Examples

```
//The following uses the restoration from the capacity scenario or customized beam scenario to the default
scenario in the 4 x 100 MHz case as an example.
//Deactivating all cells in the related sector before modification
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Setting the CSI Beam Allocation Policy to NOT_CONFIG before restoring the customized beam scenario to
the default scenario
```

```
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=0, CSIBeamAllocationPolicy=NOT_CONFIG;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=1, CSIBeamAllocationPolicy=NOT_CONFIG;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=2, CSIBeamAllocationPolicy=NOT_CONFIG;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=3, CSIBeamAllocationPolicy=NOT_CONFIG;  
//Reconfiguring the CoverageScenario, Tilt, and Azimuth parameters to identical values for all DU cells in  
the same sector  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=0, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=2, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=3, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
//Turning off the SSB Measurement Position Policy Switch when restoring the customized beam scenario to  
the default scenario  
MOD NRCELLALGOSWITCH: NrCellId=0, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-0;  
MOD NRCELLALGOSWITCH: NrCellId=1, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-0;  
MOD NRCELLALGOSWITCH: NrCellId=2, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-0;  
MOD NRCELLALGOSWITCH: NrCellId=3, MeasPolicySwitch=SSB_MEAS_POS_POLICY_SW-0;  
//Activating all cells in the related sector after modification  
ACT NRCELL: NrCellId=0;  
ACT NRCELL: NrCellId=1;  
ACT NRCELL: NrCellId=2;  
ACT NRCELL: NrCellId=3;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Run the **DSP NRDUCELLTRP** command and observe the value of **Actual Beam Scenario** in the command output. If it is consistent with the configured value, this function has been activated.

5.4.3 Network Monitoring

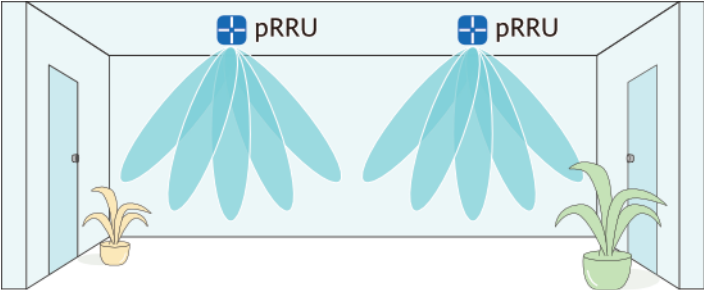
None

6 LampSite mmWave Scenario-based Beam

6.1 Principles

LampSite mmWave scenario-based beam supports the default scenario, as shown in [Figure 6-1](#). The default scenario can be configured by setting **NRDUCellTrpMmwavBeam.CoverageScenario** to **DEFAULT**.

Figure 6-1 Default scenario



"LampSite mmWave scenario-based beam" has certain requirements for RF modules. [Table 6-1](#) lists the related specifications of RF modules supporting "LampSite mmWave scenario-based beam".

Table 6-1 RF modules supporting "LampSite mmWave scenario-based beam" and related specifications

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)
DEFAULT (default scenario)	pRRU5551JR 2T2R (26 GHz)	(360°, ±80°)

 NOTE

For pRRUs of DBS5900 LampSite:

- The tilt (**NRDUCellTrpMmwavBeam.Tilt**) must be set to **255** and cannot be adjusted. (In this case, the default downtilt supported by the RF modules is used.)
- The azimuth (**NRDUCellTrpMmwavBeam.Azimuth**) must be set to **0** and cannot be adjusted. (In this case, the default azimuth 0° is used.)

6.2 Network Analysis

6.2.1 Benefits

LampSite mmWave scenario-based beam supports the default scenario, thereby ensuring user experience in LampSite base station scenarios.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

This function has no impact on the network.

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS5900 LampSite, which must be configured with the BBU5900 or BBU5910

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

pRRU5551JR (RHUB5873)

6.3.4 Cells

- The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **74, 81, 176, 180, 188, 189, 192, or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **4_1_DDDSU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).
- The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **186 or 65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).

6.3.5 Others

The **NRDUCellTrpMmwavBeam.CoverageScenario** parameter must be set to the same value for all cells in the same sector.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-2 describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

Table 6-2 Required parameters

Parameter Name	Parameter ID	Setting Notes
Coverage Scenario	NRDUCellTrpMmwav-Beam.CoverageScenario	Set this parameter to the default scenario. It is a high-risk parameter because changing the coverage scenario will lead to NR cell reestablishment and service interruption. Therefore, you are advised to perform this operation during off-peak hours.
Tilt	NRDUCellTrpMmwav-Beam.Tilt	For pRRUs of DBS5900 LampSite base stations, the tilt (NRDUCellTrpMmwav-Beam.Tilt) must be set to 255 and cannot be adjusted; in this case, the default downtilt supported by the RF module is used.
Azimuth	NRDUCellTrpMmwav-Beam.Azimuth	For pRRUs of DBS5900 LampSite base stations, the azimuth (NRDUCellTrpMmwav-Beam.Azimuth) must be set to 0 and cannot be adjusted; in this case, the default azimuth of 0° is used.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//The following uses four 100 MHz cells as an example.
//Deactivating all cells in the related sector before modification
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Configuring the CoverageScenario parameter to an identical value for all DU cells in the same sector
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;
```

```
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=2, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=3, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;  
//Activating all cells in the related sector after modification  
ACT NRCELL: NrCellId=0;  
ACT NRCELL: NrCellId=1;  
ACT NRCELL: NrCellId=2;  
ACT NRCELL: NrCellId=3;
```

Deactivation Command Examples

This function does not require deactivation configuration.

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Run the **DSP NRDUCELLTRP** command and observe the value of **Actual Beam Scenario** in the command output. If it is consistent with the configured value, this function has been activated.

6.4.3 Network Monitoring

None

7 TA Command Delivery at a Short Interval

7.1 Principles

During uplink timing, the gNodeB measures the signals sent from a UE to obtain a timing offset and delivers the timing offset to the UE through a timing advance (TA) command. The UE then adjusts the time to transmit signals based on this command. The entire process ensures uplink synchronization between the UE and the gNodeB. Angle of arrivals (AoAs) vary with beams, and therefore the timing offset of a UE may also be significantly different among beams. In this situation, the gNodeB also needs to deliver TA commands to align uplink timing during beam switching. A shorter interval for TA command delivery means more timely uplink timing alignment. The function of TA command delivery at a short interval is controlled by the **SHORT_TAC_PERIOD** option of the **NRDUCellAlgoSwitch.ShortTaCommandPeriodSW** parameter.

- If this option is selected, the gNodeB delivers TA commands at a short interval during beam switching. After the switching, the gNodeB delivers TA commands at the original interval.
- If this option is deselected, the interval for TA command delivery remains unchanged during beam switching.

For more information about TA commands, see section 4.2 "Transmission timing adjustments" in 3GPP TS 38.213 V16.1.0.

7.2 Network Analysis

7.2.1 Benefits

During beam switching, the gNodeB delivers TA commands at a short interval to promptly align uplink timing, reducing the service drop rate caused by inaccurate uplink timing while improving user experience.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

During beam switching, the gNodeB frequently delivers TA commands, consuming an increased number of air interface resources. As a result, the downlink MCS index may be reduced, compromising the transmission efficiency.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

DBS5900 LampSite, which must be configured with the BBU5900 or BBU5910

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5213, HAAU5222, HAAU5323, HAAU5233w, HAAU5235w, pRRU5551JR (RHUB5873)

7.3.4 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

[Table 7-1](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Short TA Command Period Switch	NRDUCellAlgoSwitch.ShortTaCommandPeriodSW	SHORT_TAC_PERIOD	Set this parameter to its recommended value.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Selecting the SHORT_TAC_PERIOD option  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ShortTaCommandPeriodSW=SHORT_TAC_PERIOD-1;
```

Optimization Command Examples

None

Deactivation Command Examples

```
//Deselecting the SHORT_TAC_PERIOD option  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ShortTaCommandPeriodSW=SHORT_TAC_PERIOD-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

None

7.4.3 Network Monitoring

None

8 Flexible Dense Beam

Flexible Dense Beam is an enhancement of original basic CSI-RS for beam measurement (CSI-RS for BM) beams and includes beam densification and flexible sweeping. In beam densification, the number of CSI-RS for BM beams is increased to obtain more accurate CSI-RS for BM beams targeted at UEs. In this way, coverage and throughput gains are obtained. In flexible sweeping, unnecessary CSI-RS for BM beam sweeping is reduced to offset the increase in time-frequency resource overheads caused by an increase in the number of beams.

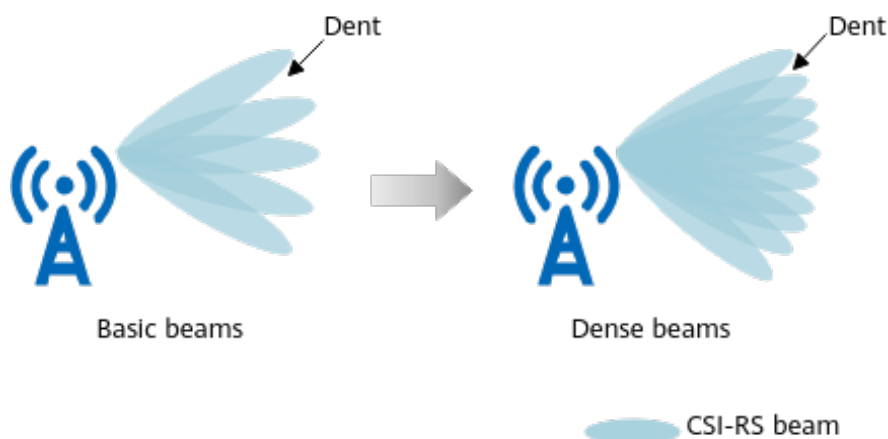
8.1 Principles

The Flexible Dense Beam function involves beam densification and flexible sweeping. The two sub-functions take effect simultaneously by selecting the **FLEXIBLE_DENSE_BEAM_SW** option of the **NRDUCellTrpMmwavBeam.BeamPerformanceSw** parameter.

Beam Densification

The hybrid beamforming architecture is used in high frequency bands. Digital weighting is performed on the basis of analog beams to obtain the final beams. Analog beams are discrete and discontinuous, and there are dents between discrete analog beams. To address this issue, the beam densification function is introduced. As shown in [Figure 8-1](#), this function increases the number of CSI-RS for BM analog beams to obtain more accurate beams targeted at UEs and reduce the dents between discrete analog beams, thereby improving signal quality and obtaining throughput gains.

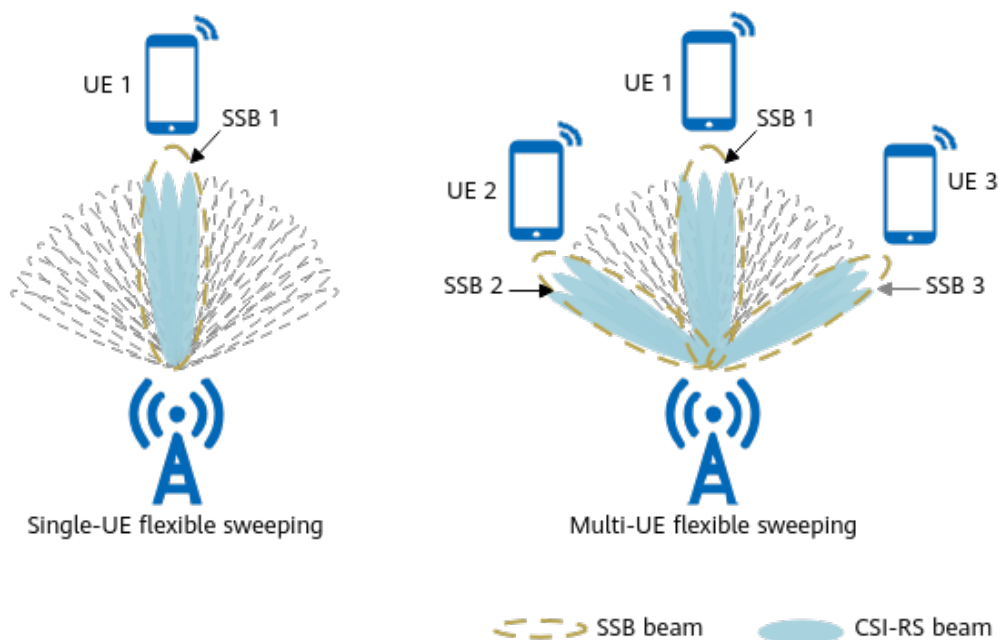
Figure 8-1 Example of beam densification



Flexible Sweeping

After the beam densification function is introduced, the number of CSI-RS for BM analog beams increases, which increases the time-frequency resource consumption. To reduce the foregoing time-frequency resource consumption, the flexible sweeping function is introduced. As shown in [Figure 8-2](#), a base station determines the range of CSI-RS for BM beams to be sent based on a serving SSB beam of a UE. If there is no UE in connected mode on the SSB beam, a corresponding CSI-RS for BM beam does not need to be swept or sent, therefore, time-frequency resource consumption of beam sweeping is reduced.

Figure 8-2 Example of flexible sweeping



8.2 Network Analysis

8.2.1 Benefits

After the Flexible Dense Beam function is enabled, the basic CSI-RS for BM beams are further enhanced to form flexible and dense beams. Compared with basic CSI-RS for BM beams, flexible and dense beams can target at UEs more accurately. It is recommended that Flexible Dense Beam be enabled when UEs are evenly distributed in a cell and the proportion of UEs whose channel quality reaches the peak value is less than 5%. With this function enabled, the **Cell Downlink Average Throughput (DU)** or **User Downlink Average Throughput (DU)** can increase by 3% to 10%, and the **Cell Uplink Average Throughput (DU)** or **User Uplink Average Throughput (DU)** can increase by 3% to 15%. The more densely UEs are distributed in a cell and the higher the proportion of UEs whose channel quality reaches the peak value in the cell, the lower the positive gains.

In non-recommended scenarios, the Flexible Dense Beam function still takes effect. However, the carrier resources of the cell cannot be fully used. As a result, the positive gains decrease or no positive gains are generated.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

In non-recommended scenarios, enabling Flexible Dense Beam may slightly decrease the **Cell Downlink Average Throughput (DU)** or **User Downlink Average Throughput (DU)** when the average downlink MCS index in a cell is greater than 26.

8.3 Requirements

8.3.1 Licenses

None

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	CSI-RS for BM in SSB slots	SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	When the CSI-RS for BM in SSB slots function is disabled, the Flexible Dense Beam function cannot be enabled.

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Configuration of the CSI-RS for BM measurement period	NRDUCellCsirs.CsiRsBeamMeasPeriod	<i>Channel Management</i>	When the Flexible Dense Beam function is enabled, the NRDUCellCsirs.CsiRsBeamMeasPeriod parameter cannot be set to SLOT160 .

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Configuration of the CSI-RS for BM measurement period	NRDUCellCsirs.CsiRsBeamMeasurementPeriod	<i>Channel Management</i>	The Flexible Dense Beam function involves beam densification and flexible sweeping. The two sub-functions take effect simultaneously by selecting the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter. After beam densification takes effect, the resource consumption of beam sweeping increases with the number of CSI-RS for BM beams. In addition, flexible sweeping can reduce the resource consumption in a cell serving a small number of UEs, but significantly increase the resource consumption in a cell serving a large number of UEs. As prolonging the beam sweeping period can reduce resource consumption, you are advised to reconfigure the CSI-RS for BM measurement period when the Flexible Dense Beam function is enabled. • In eMBB scenarios, you are advised to set the NRDUCellCsirs.CsiRsBeamMeasurementPeriod parameter to SLOT320. • In FWA scenarios, you are advised to set the NRDUCellCsirs.CsiRsBeamMeasurementPeriod parameter to SLOT640.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Multi-beam FDM	VOL_BASED_BEAM_MULTIPLEX_SW option of the NRDUCellAlgorithmSwitch.BeamMultiplexSwitch parameter	<i>MIMO (TDD)</i>	After the Flexible Dense Beam function takes effect, the number of data transmission beams may increase and the beam multiplexing degree may decrease. Therefore, it is recommended that multi-beam FDM be enabled together with Flexible Dense Beam to improve the beam multiplexing degree. Otherwise, the Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU) may decrease due to a decrease in the beam multiplexing degree.
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	- In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), the switch for Flexible Dense Beam can be turned on, but the function does not take effect. - In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_104(Wide Beam Scenario)), the gains of Flexible Dense Beam may decrease because coverage scopes of SSB and CSI-RS beams may change.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgorithmSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	After Flexible Dense Beam is enabled together with mmWave PUSCH MU-MIMO, the cell throughput changes because the serving beams of UEs and the measured analog beam interference between UEs change.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5213, HAAU5222, HAAU5323, HAAU5233w, HAAU5235w

8.3.4 Cells

When the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**, the Flexible Dense Beam function can be switched on but will not take effect.

In other slot configuration scenarios, the preceding restrictions do not apply.

8.3.5 Others

The Flexible Dense Beam function must be enabled or disabled simultaneously for all activated cells in the same sector.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

[Table 8-1](#) describes the required parameters.

Table 8-1 Required parameters

Parameter Name	Parameter ID	Setting Notes
Beam Performance Switch	NRDUCellTrpMmwav-Beam.BeamPerformanceSw	The FLEXIBLE_DENSE_BEAM_SW option specifies whether to enable Flexible Dense Beam. Set this option to its recommended value. High-risk parameter: Changing the setting of this option will cause a high-frequency cell to automatically restart. Therefore, you are advised to perform this operation during off-peak hours.
CSI-RS Beam Measurement Period	NRDUCellCsirs.CsiRsBeamMeasPeriod	- In eMBB scenarios, the value SLOT320 is recommended. - In FWA scenarios, the value SLOT640 is recommended. High-risk parameter: Changing the setting of this parameter will cause a high-frequency cell to automatically restart. Therefore, you are advised to perform this operation during off-peak hours.
Beam Multiplex Switch	NRDUCellAlgoSwitch.BeamMultiplexSwitch	It is recommended that the VOL_BASED_BEAM_MULTIPLEX_SW option be selected to improve the beam multiplexing degree when Flexible Dense Beam is enabled.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	For burst services, it is recommended that the PUSCH_DTX_AGG_LVL_ADAPT_SW option be selected to prevent multi-beam FDM from being ineffective due to insufficient PDCCH beams.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Deactivating all cells in the corresponding sector before the Flexible Dense Beam function is enabled
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Enabling the Flexible Dense Beam function simultaneously for all activated cells in the same sector
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-1;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=1, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-1;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=2, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-1;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=3, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-1;
//Activating all cells in the corresponding sector after enabling the Flexible Dense Beam function
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
```

Optimization Command Examples

```
//Setting the CsiRsBeamMeasPeriod parameter to reduce the increase in the consumption caused by the
increase in the number of beams in eMBB scenarios when Flexible Dense Beam is enabled
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsBeamMeasPeriod=SLOT320;
//(Recommended) Setting the CsiRsBeamMeasPeriod to SLOT640 in FWA scenarios where the measurement
period of CSI-RS for BM beams can be longer than that in eMBB scenarios because UEs are stationary
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiRsBeamMeasPeriod=SLOT640;
//(Recommended) Enabling multi-beam FDM when Flexible Dense Beam is enabled to improve the beam
multiplexing degree and enabling PUSCH-DTX-based aggregation level adaptation for burst services to
prevent multi-beam FDM from being ineffective due to limited PDCCH beams
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, BeamMultiplexSwitch=VOL_BASED_BEAM_MULTIPLEX_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-1;
```

Deactivation Command Examples

```
//Deactivating all cells in the corresponding sector before the Flexible Dense Beam function is disabled
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
//Disabling the Flexible Dense Beam function simultaneously for all activated cells in the same sector
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=0, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=1, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=2, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-0;
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpId=3, BeamPerformanceSw=FLEXIBLE_DENSE_BEAM_SW-0;
//Activating all cells in the corresponding sector after disabling the Flexible Dense Beam function
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

After the Flexible Dense Beam function is enabled, you can monitor the event block "PERIOD_PRIVATE_UE_MEASUREMENT" in external CHR logs and check the value of **CsiRs_Beam_ID** under **UEMeasInfo > InstantMeasItems > CsiRs_Beam**. If there is a **CsiRs_Beam_ID** whose value is greater than or equal to 64, Flexible Dense Beam has taken effect.

8.4.3 Network Monitoring

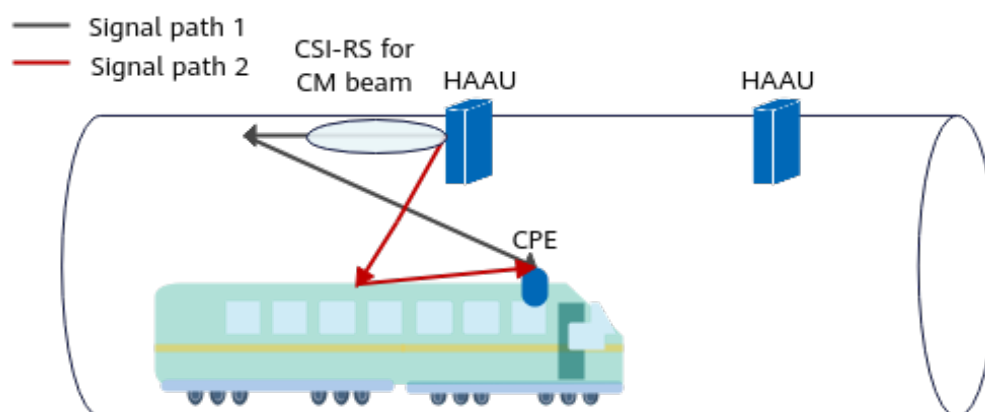
Observe whether the following KPIs change as expected in [8.2.1 Benefits](#):

- **User Uplink Average Throughput (DU)**
- **Cell Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**
- **Cell Downlink Average Throughput (DU)**

9 Dynamic Beam Allocation

As shown in [Figure 9-1](#), a CPE is in mobile backhaul scenarios with high-frequency deployment such as subway tunnels. When **NRDUCellTrpMmwavBeam.CoverageScenario** is set to **SCENARIO_103**, the direction of the CSI-RS for channel measurement (CSI-RS for CM) beam is fixed, and the CPE in the front or rear of the train enters the handover area, the multipath effect between the CPE and the HAAU is severe, frequency-selective fading is likely to occur, the multipath delay spread is likely to exceed the cyclic prefix (CP), and inter-symbol interference (ISI) is generated. As a result, the downlink UE throughput decreases significantly when the CPE enters the handover area.

Figure 9-1 Background of dynamic beam allocation



Against this backdrop, Huawei introduces dynamic beam allocation. This function increases the downlink UE throughput by mitigating the multipath effect when the CPE in the front or rear of the train enters a handover area. This function takes effect only for the CPE in the front or rear of the train in mobile backhaul scenarios with high-frequency deployment such as subway tunnels. A maximum of two CPEs can be deployed in the front or rear of the train, and there can be up to two CPEs in the front or rear of the train in a sector.

NOTE

CPEs here refer to train access units (TAUs) deployed in mobile backhaul scenarios such as subway tunnels.

Due to the particularity of HAAU and CPE deployment in mobile backhaul scenarios, an HAAU covers either the CPE in the front of or that in the rear of the train. Therefore, which CPE an HAAU serves has been determined during deployment planning. The principles of dynamic beam allocation are the same for the CPEs in the front and rear of the train. The following part uses the CPE in the front of the train as an example to describe the principles of dynamic beam allocation.

9.1 Principles

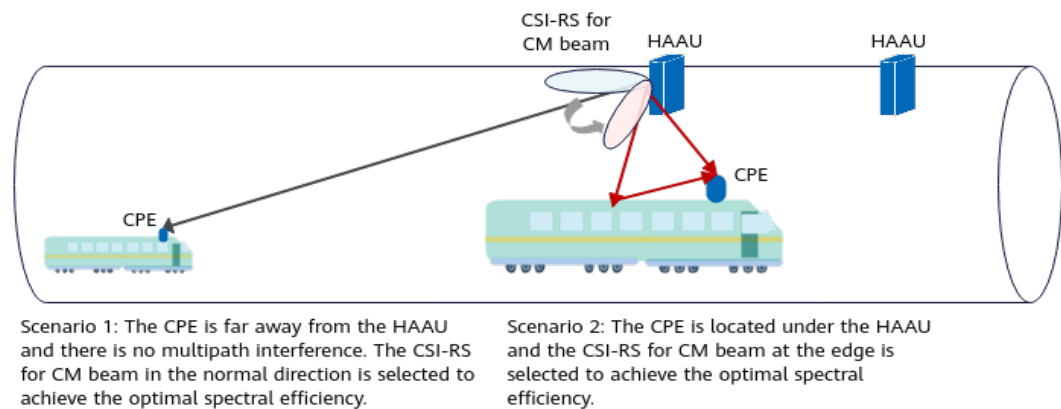
When the CPE is in mobile backhaul scenarios with high-frequency deployment such as subway tunnels, dynamic beam allocation is enabled in the serving cell if all of the following conditions are met:

- The **DYNAMIC_BEAM_ALLOC_SW** option of the **NRDuCellMobileBhAlgo.MobileBackhaulAlgoSw** parameter is selected for the serving cell.
- Each activated carrier of the serving cell can serve only one CPE.
- After the serving cell receives SRSs from the CPE for three consecutive times, each of the calculated maximum multipath delay spread is greater than the value of **NRDuCellMobileBhAlgo.DynBeamAllocDelayDiffThld**.

This condition is met as long as the multipath delay spread calculated after the serving cell receives the SRSs reported by one CPE meets the preceding threshold requirement.

As shown in [Figure 9-2](#), after dynamic beam allocation is enabled, the gNodeB calculates the spectral efficiency of the CSI-RS for CM beam based on the channel quality measured and reported by the CPE in the front of the train and selects the CSI-RS for CM beam with the optimal spectral efficiency for the CPE. After this function is enabled, the gNodeB can dynamically adjust the direction of CSI-RS for CM beams based on the location of the CPE in the front of the train.

Figure 9-2 Dynamic beam allocation



9.2 Network Analysis

9.2.1 Benefits

It is recommended that dynamic beam allocation be enabled when the CPE is in mobile backhaul scenarios with high-frequency deployment such as subway tunnels. After this function is enabled, the average downlink UE throughput increases when the CPE is located in the overlapping area of two cells.

NRDuCellMobileBhAlgo.DynBeamAllocDelayDiffThld has the following impacts on gains:

- A smaller value of this parameter results in a higher probability of enabling dynamic beam allocation and higher gains. However, if this threshold is set to an excessively small value, CSI-RS for BM beam sweeping resources may be wasted. As a result, the average downlink UE throughput may decrease.
- A higher threshold results in a lower probability of enabling dynamic beam allocation and lower gains.

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

This function has no impact on the network.

9.3 Requirements

9.3.1 Licenses

None

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	When dynamic beam allocation is enabled, the NRDUCellTrpMmwavBeam.CoverageScenario parameter must be set to SCENARIO_103 .

9.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Low Latency High Reliability Level 2	LOW_LAT_HIGH_RELIAB_LEVEL2_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	When dynamic beam allocation is enabled, the LOW_LAT_HIGH_RELIAB_LEVEL2_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter must be deselected.
High-frequency TDD	mmWave level-1 low jitter and high reliability	L_JIT_H_RELIAB_MMW_LVL1_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	When dynamic beam allocation is enabled, the L_JIT_H_RELIAB_MMW_LVL1_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter must be deselected.

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH grouping adaptation enhancement for inter-FR CA	PUCCH_GROUP_ADAPT_SW and PUCCH_GROUP_ADAPT_ENH_SW options of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Dynamic beam allocation does not take effect for UEs with PUCCH grouping adaptation enhancement for inter-FR CA in effect.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All RF modules that support the single-beam scenario of mmWave 3D coverage pattern support this function. For details, see [Table 5-1](#) in [5.1 Principles](#).

9.3.4 Cells

NRDUCell.SlotAssignment must be set to **4_1_DDDSU**.

9.3.5 Others

Dynamic beam allocation must be enabled or disabled for all activated cells in the same sector at the same time. Otherwise, this function may not take effect.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

[Table 9-1](#) describes the required parameters.

Table 9-1 Required parameters

Parameter Name	Parameter ID	Setting Notes
Mobile Backhaul Algorithm Switch	NRDuCellMobileBhAlgo.MobileBackhaulAlgoSw	It is recommended that the DYNAMIC_BEAM_ALLOC_SW option be selected when the CPE is in mobile backhaul scenarios with high-frequency deployment such as subway tunnels.
Dynamic Beam Allocation Delay Diff Thld	NRDuCellMobileBhAlgo.DynBeamAllocDelayDiffThld	Use the recommended value. This parameter takes effect only when the DYNAMIC_BEAM_ALLOC_SW option of the NRDuCellMobileBhAlgo.MobileBackhaulAlgoSw parameter is selected.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling dynamic beam allocation simultaneously for all activated cells in the same sector
MOD NRDUCELLMOBILEBHALGO: NrDuCellId=0, MobileBackhaulAlgoSw=DYNAMIC_BEAM_ALLOC_SW-1;
//Configuring Dynamic Beam Allocation Delay Diff Thld
MOD NRDUCELLMOBILEBHALGO: NrDuCellId=0, DynBeamAllocDelayDiffThld=15;
```

Deactivation Command Examples

```
//Disabling dynamic beam allocation simultaneously for all activated cells in the same sector
MOD NRDUCELLMOBILEBHALGO: NrDuCellId=0, MobileBackhaulAlgoSw=DYNAMIC_BEAM_ALLOC_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

None

9.4.3 Network Monitoring

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
- Step 2** In the navigation tree on the left of the displayed **Signaling Trace Management** tab page, choose **NR > Cell Performance Monitoring > General Throughput Monitoring**.
- Step 3** Monitor the increase in the total traffic volume in the cell handover area.

 NOTE

In mobile backhaul scenarios such as subway tunnels, a cell covers only one train at a time and serves one CPE. Therefore, the increase in UE throughput can be observed based on cell traffic volume indicators.

----End

10 Optimal Selection of Narrow Beams for Access Channels

Optimal selection of narrow beams for access channels is an enhancement to beamforming during random access. Traditionally, wide-beam-based access is performed by a UE. However, with this function, after the gNodeB evaluates the direction of the UE that initiates access during message (Msg1) reception on the PRACH, the optimal narrow beam is selected from the predefined set of narrower beams with higher gains. This beam is used to send and receive subsequent access messages (Msg2 to Msg5), improving the coverage performance of UE during access.

10.1 Principles

When a UE initiates random access, the gNodeB can only obtain the SSB wide beam used by the UE but cannot predict the accurate UE direction. In this case, Msg1 to Msg5 are received and sent using wide beams with the same width as the SSB wide beam during the random access, resulting in low beam gains.

In this context, optimal selection of narrow beams for access channels is introduced. This function is controlled by the **ACC_CHN_OPT_NARROW_BEAM_SEL_SW** option of the **NRDUCellTrpMmwavBeam.BeamPerformanceSw** parameter.

After this function is enabled, the gNodeB adjusts analog beamforming during PRACH reception to determine the UE direction. After the UE's PRACH signals are detected, the gNodeB selects an optimal beam from the set of candidate analog narrow beams based on the UE direction, and uses corresponding analog beamforming in subsequent reception and transmission of access messages (Msg2 to Msg5). In this way, the beam gains of Msg2 to Msg5 are improved, thereby improving the overall coverage performance of the access channel. If the UE's PRACH signal cannot be detected, the UE direction cannot be determined and optimal selection of narrow beams for access channels cannot take effect.

10.2 Network Analysis

10.2.1 Benefits

This function uses narrower beams with higher gains to send and receive access channel messages. It increases the signal-to-noise ratio (SNR) of the access channel, improves the coverage performance of cells by 2 dB to 3 dB on average during access, and allows UEs to access the network at a longer distance or in a position with a higher path loss. The gains depend on the UE location.

- When a UE is at the vertical coverage edge of an SSB beam, the gain is relatively small or no gain is generated.
- When a UE is within the vertical coverage scope of an SSB beam, the gain is relatively large.

In the following scenarios, optimal selection of narrow beams for access channels provides no gains:

- UEs whose coverage is not limited during access
- UEs whose PRACH signals cannot be detected
- For 4T4R RF modules, the **NRDUCellPrach.PuschPrachFdmSwitch** parameter is set to **BASIC** or **FULL_BUFFER**, and there are UEs performing large-packet services in the sector.

After optimal selection of narrow beams for access channels is enabled and the coverage performance during access is improved, the following situation may occur: UEs with limited coverage during access can successfully access the network. However, when downlink data is transmitted after their access, the coverage of short PUCCH is limited, resulting in HARQ feedback reception failures. Therefore, it is recommended that adaptive HARQ-ACK feedback on downlink transmission (controlled by the **ACK_NACK_REPORTING_PUSCH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter) be enabled together with optimal selection of narrow beams for access channels to improve the coverage performance of both access and data channels. For details about adaptive HARQ-ACK feedback on downlink transmission, see *Downlink Scheduling*.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

- If UEs at the cell edge access a cell, the values of **RRC Setup Success Rate (CU)** and **QoS Flow Setup Success Rate (CU)** may increase.
- The values of **N.User.RRCCConn.Max**, **N.User.RRCCConn.Avg**, **N.User.NsaDc.PSCell.Avg**, **N.User.RRCCConn.Active.Max**, and **N.User.RRCCConn.Active.Avg** may increase.
- If UEs that originally failed to access the network can access the network after this function is enabled, the uplink and downlink PRB usages of the cell may increase in light-load scenarios and decrease in heavy-load scenarios.
 - Uplink PRB usage: **Uplink Resource Block Utilizing Rate**
 - Downlink PRB usage: **Downlink Resource Block Utilizing Rate**
- If UEs that originally failed to access the network can access the network after this function is enabled, the values of **N.PDCP.Vol.UL.TrfSDU.Tx**, **N.PDCP.Vol.DL.TrfSDU.Rx**, **N.ThpVol.DL**, and **N.ThpVol.UL** may increase.

- If UEs that originally failed to access the network can access the network after this function is enabled, the average uplink and downlink MCS indexes and ranks of UEs slightly decrease.
 - Average uplink MCS index:

$$\text{Average uplink MCS index} = (0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})$$
 - Average downlink MCS index:

$$\text{Average downlink MCS index} = (0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PDSCH.MCS.28}) / (\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \dots + \text{N.ChMeas.PDSCH.MCS.28})$$
 - Average uplink rank:

$$\text{Average uplink rank} = (1 \times \text{N.PUSCH.InitTbUL.Rank1} + 2 \times \text{N.PUSCH.InitTbUL.Rank2}) / (\text{N.PUSCH.InitTbUL.Rank1} + \text{N.PUSCH.InitTbUL.Rank2})$$
 - Average downlink rank:

$$\text{Average downlink rank} = (1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4}) / (\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})$$
- The average uplink and downlink cell throughputs fluctuate.
 - Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
 - Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- The average uplink and downlink UE throughputs fluctuate. When a large number of CEUs access the network, the average uplink and downlink UE throughputs may decrease.
 - Average uplink UE throughput: **User Uplink Average Throughput (DU)**
 - Average downlink UE throughput: **User Downlink Average Throughput (DU)**
- If UEs that originally failed to access the network can access the network after this function is enabled, the PDCCH CCE usage (**N.CCE.Used.Avg/N.CCE.Avail.Avg**) may increase.

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	Optimal selection of narrow beams for access channels can be enabled only after the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to DEFAULT .
High-frequency TDD	PRACH configuration index	NRDUCellPrach.PrachConfigurationIndex	None	Optimal selection of narrow beams for access channels can be enabled only after the NRDUCellPrach.PrachConfigurationIndex parameter is set to 74 or 81 .

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUSCH and PRACH FDM	NRDUCellPrach.PuschPrachFdmSwitch	<i>Uplink Scheduling</i>	For 4T4R modules, optimal selection of narrow beams for access channels does not take effect if the NRDUCellPrach.PuschPrachFdmSwitch parameter is set to BASIC or FULL_BUFFER and there are speed turbo UEs or MNT UEs in the sector.

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5323, HAAU5233w, HAAU5235w

10.3.4 Cells

- **NRDUCell.SlotAssignment** must be set to **4_1_DDDSU**.
- **NRDUCellTrp.TxRxMode** must be set to **4T4R**.

10.3.5 Others

Optimal selection of narrow beams for access channels must be enabled for all activated cells under the same sector. Otherwise, this function cannot take effect.

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

[Table 10-1](#) describes the required parameters.

Table 10-1 Required parameters

Parameter Name	Parameter ID	Option	Setting Notes
Beam Performance Switch	NRDUCellTrpMmwavBeam.BeamPerformanceSw	ACC_CHN_ OPT_NARR OW_BEAM_ SEL_SW	Use the recommended setting.

10.4.1.2 Using MML Commands

Before using MML commands, refer to [10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling optimal selection of narrow beams for access channels for all activated cells in the same sector  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=0,  
BeamPerformanceSw=ACC_CHN_OPT_NARROW_BEAM_SEL_SW-1;
```

Deactivation Command Examples

```
//Disabling optimal selection of narrow beams for access channels for all activated cells in the same sector  
MOD NRDUCELLTRPMMWAVBEAM: NrDuCellTrpld=0,  
BeamPerformanceSw=ACC_CHN_OPT_NARROW_BEAM_SEL_SW-0;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

After optimal selection of narrow beams for access channels is enabled, if the number of online users in a cell and the RACH access success rate increase, this function has taken effect.

- Number of online users: **N.User.RRCCConn.Max**, **N.User.RRCCConn.Avg**, **N.User.NsaDc.PSCell.Avg**, **N.User.RRCCConn.Active.Max**, and **N.User.RRCCConn.Active.Avg**
- RACH access success rate:
 - RACH RAR Response Rate of Contention = **N.RA.Contention.Resp/N.RA.Contention.Att** x 100%
 - RACH RAR Response Rate of Dedicated = **N.RA.Dedicated.Resp/N.RA.Dedicated.Att** x 100%

10.4.3 Network Monitoring

None

11

Antenna Large Physical Downtilt Mode

11.1 Principles

In stadium scenarios where RF modules are installed at heights with a short coverage distance, the RF modules may require large downtilts according to the network plan. However, mmWave RF modules support an adjustable range of mechanical downtilt from -20° to $+20^{\circ}$ to meet heat dissipation requirements in the working environment. If the mechanical downtilt exceeds the adjustable range, a high-temperature alarm will be reported and therefore RF modules cannot work properly. To meet the requirements in the preceding scenarios, the antenna large physical downtilt mode is introduced to allow RF modules to be horizontally installed or installed with a large physical downtilt. For details, see [Figure 11-1](#) and [Figure 11-2](#).

Figure 11-1 Horizontal installation

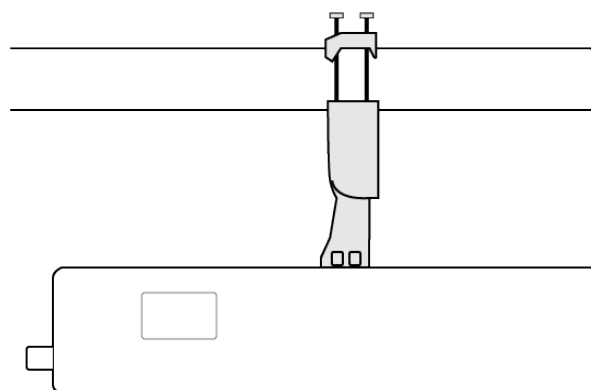
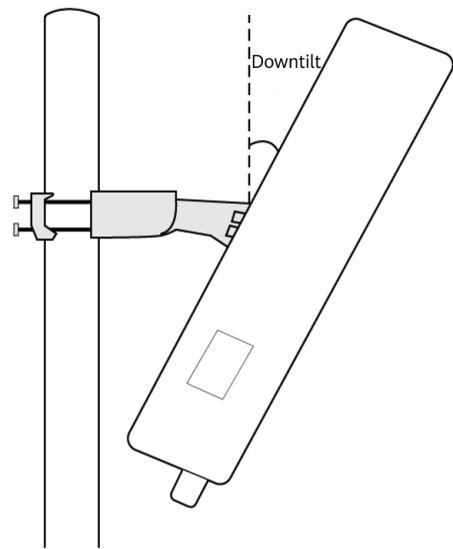


Figure 11-2 Installation with a large physical downtilt



 NOTE

Horizontal installation refers to the case that RF modules are installed horizontally.
A large physical downtilt indicates the downtilt of an RF module when its mechanical downtilt exceeds the adjustable range from -20° to $+20^{\circ}$.

The antenna large physical downtilt mode can reduce the power consumption of RF modules by shutting down partial antenna elements of the RF modules. This helps reduce the heat generated when RF modules are horizontally installed or installed with a large physical downtilt to meet the heat dissipation requirements in the working environment. This function is controlled by the **ANT_LARGE_PHY_TILT_MODE_SW** option of the **RRUEXTINFO.RRUOPTLFEATSW** parameter.

The vertical 3 dB beamwidth increases after the antenna large physical downtilt mode is enabled. The beamwidth is related to the configured coverage scenario (specified by the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter) of mmWave 3D coverage pattern. For details, see [Table 11-1](#).

Table 11-1 Beamwidths of RF modules before and after enabling the antenna large physical downtilt mode

Scenario ID	RF Module	Before Function Enabling (Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	After Function Enabling (Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)
DEFAULT (default scenario)	HAAU5233w 4T4R (26 GHz)	(110°, 18°)	(110°, 25°)
DEFAULT (default scenario)	HAAU5235w 4T4R (28 GHz)	(110°, 18°)	(110°, 25°)

Scenario ID	RF Module	Before Function Enabling (Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)	After Function Enabling (Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)
SCENARIO_101 (capacity scenario)	HAAU5233w 4T4R (26 GHz)	(110°, 18°)	(110°, 25°)
SCENARIO_101 (capacity scenario)	HAAU5235w 4T4R (28 GHz)	(110°, 18°)	(110°, 25°)
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5233w 4T4R (26 GHz)	- SSB beam pattern 1: (7°, 6°) for a single beam - SSB beam pattern 2: (14°, 6°) for a single beam - SSB beam pattern 3: (28°, 6°) for a single beam	- SSB beam pattern 1: (7°, 12°) for a single beam - SSB beam pattern 2: (14°, 12°) for a single beam - SSB beam pattern 3: (28°, 12°) for a single beam
CUSTOMIZED_BEAM_SCENARIO_106 (customized beam scenario)	HAAU5235w 4T4R (28 GHz)	- SSB beam pattern 1: (7°, 6°) for a single beam - SSB beam pattern 2: (14°, 6°) for a single beam - SSB beam pattern 3: (28°, 6°) for a single beam	- SSB beam pattern 1: (7°, 12°) for a single beam - SSB beam pattern 2: (14°, 12°) for a single beam - SSB beam pattern 3: (28°, 12°) for a single beam

11.2 Network Analysis

11.2.1 Benefits

After the antenna large physical downtilt mode is enabled, the heat generated by RF modules when they are horizontally installed or installed with a large physical downtilt can be reduced. This helps RF modules to meet heat dissipation requirements in the working environment and prevents high-temperature alarms from being generated on RF modules. In this way, RF modules now can be

horizontally installed or installed with a large physical downtilt in scenarios such as stadiums.

11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

The antenna large physical downtilt mode has the following impacts on the network:

- The maximum EIRP of the cell served by the RF module is 6 dB lower than the configured maximum EIRP (specified by the **NRDUCellTrp.MaxEirp** parameter). In other words, the cell coverage range shrinks accordingly.
- During the period from when the function is enabled to when it takes effect, instantaneous fluctuations occur in the KPIs related to access, handover, and service drop of the cell served by the RF module.

11.3 Requirements

11.3.1 Licenses

None

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	The antenna large physical downtilt mode cannot be enabled in the following scenarios: - Coverage scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_102(Coverage Scenario)) - Single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)) - Wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario))

11.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Beam-based power control	WRC19_EIRP_CONTROL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	After the antenna large physical downtilt mode is enabled, the maximum EIRP of the cell decreases by 6 dB. As a result, the power control value of beam-based power control may decrease.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Grid-specific power control in the smart power lock	AVERAGE_3D EIRP_CONTR OL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	After the antenna large physical downtilt mode is enabled, the maximum EIRP of the cell decreases by 6 dB. As a result, the power control value of grid-specific power control in the smart power lock may decrease.
High-frequency TDD	Power control for the upper hemisphere of TRPs	UP_HEMISPHERE_TRP_POWER_CTRL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter	None	After the antenna large physical downtilt mode is enabled, the maximum EIRP of the cell decreases by 6 dB. As a result, the power control value of upper hemisphere TRP power control may decrease.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	After the antenna large physical downtilt mode is enabled, the vertical 3 dB beamwidth is expanded in the following mmWave 3D coverage pattern scenarios: • Default scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to DEFAULT(Default Scenario)) • Capacity scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_101(Capacity Scenario)) • Customized beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to CUSTOMIZED_BEAM_SCENARIO_106(Customized Beam Scenario))

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5233w, HAAU5235w

11.3.4 Others

None

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

Table 11-2 describes the required parameter.

Table 11-2 Required parameters

Parameter Name	Parameter ID	Option	Setting Notes
RRU Optional Feature Switch	RRUEXTINFO. <i>RRUOPTLFEATSW</i>	ANT_LARGE_PHY_TILT_MODE_SW	It is recommended that this option be selected if the mechanical downtilt of an RF module exceeds the adjustment range (–20° to +20°). Otherwise, an alarm indicating RF Unit Temperature Unacceptable will be generated.

11.4.1.2 Using MML Commands

Before using MML commands, refer to [11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the function of antenna large physical downtilt mode (in a scenario where RRU extension information has not been added)  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUOPTLFEATSW=ANT_LARGE_PHY_TILT_MODE_SW-1;
```



```
//Enabling the function of antenna large physical downtilt mode (in a scenario where RRU extension  
information has been added)  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUOPTLFEATSW=ANT_LARGE_PHY_TILT_MODE_SW-1;
```

Deactivation Command Examples

```
//Disabling the function of antenna large physical downtilt mode  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUOPTLFEATSW=ANT_LARGE_PHY_TILT_MODE_SW-0;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

Run the **DSP RRUEXTINFO** command to query the effective status of the antenna large physical downtilt mode switch. If the antenna large physical downtilt mode is displayed to be enabled, this function has taken effect.

11.4.3 Network Monitoring

None

12 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

13 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

14 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

15 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- *Beam Management*
- *Channel Management*
- *Power Control*
- *MIMO (TDD)*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Random Access Control*
- *Carrier Aggregation*
- *License Control Item Lists*
- *Standards Compliance*
- *mmWave Super Coverage (High-Frequency TDD)*
- *Intelligent Aggregate Scheduling (High-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*